

**Machine Learning-Based Prediction of AI-Induced Career
Anxiety Among University Students.**

Final Year Design Project

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FINAL YEAR DESIGN PROJECT REPORT

This Report Presented in Partial Fulfillment of the Requirements for
the **Degree of Bachelor of Science in Computer Science and
Engineering**

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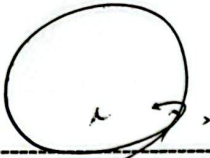
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APPROVAL

This project, titled **Machine Learning-Based Prediction of AI-Induced Career Anxiety Among University Students**, submitted by Md. Mohsin, ID No: **0242220005101539**, and Md. Isfak Sikder, ID No: **0242220005101537** to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on **15-16 September, 2026**.

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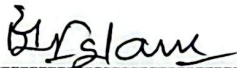
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We hereby declare that this project has been done by us under the supervision of **Md. Shahriar Shakil, Lecturer**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere for the award of any degree or diploma.

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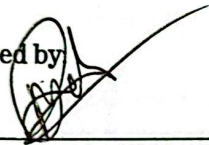


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ABSTRACT

The rapid rise of AI tools in academic and professional environments has increased concerns among university students about their future career prospects, a phenomenon increasingly referred to in the literature as AI-related career anxiety. Although previous studies have primarily used descriptive, correlational, and structural equation modeling approaches, relatively little research has examined this issue through individual-level, held-out predictive evaluation, particularly in the context of Bangladeshi universities. This study addresses this gap by treating AI-related career anxiety as a binary supervised classification problem. A survey dataset containing 3,156 respondents was used, from which an analytical sample of 2,036 second-, third-, and fourth-year students was selected. Seventeen predictors covering demographic characteristics, perceptions of AI, and engineered measures of AI-tool usage were developed and evaluated using a leak-safe preprocessing pipeline, an 80/20 stratified train-test split, and stratified 10-fold cross-validation across four cohorts: Overall, Public, Private, and Daffodil, with Daffodil considered a nested subset of the Private cohort. Ten classifiers were compared, together with approaches for handling class imbalance, LASSO-based feature selection, SHAP-based explainability, feature-set ablation, and ensemble learning. Using the frozen Full17 feature representation, Gradient Boosting produced the highest mean Test F1 score (0.7901) across the cohorts. However, predictive separability differed substantially between cohorts: the Daffodil cohort achieved the highest ROC-AUC (0.7402), whereas the Public cohort performed close to chance level (0.4846). Two predictors, `ai_future_perspective` and `ai_takeover_time`, remained stable across all LASSO subsets, while seven predictors consistently appeared among the most influential SHAP features across cohorts. Overall, the findings provide an interpretable and cohort-aware machine-learning framework that can serve as a methodological foundation for future student-support applications, while avoiding claims of causal relationships or clinical validity.

Keywords: *Artificial Intelligence, Career Anxiety, Machine Learning, Explainable AI, SHAP, LASSO, University Students, Bangladesh*

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Chapter 1

Introduction

The current chapter addresses the problem of anxiety related to AI and career choices among university students, specifies the motivation and goals of the study, gives a brief methodology description, and describes the structure of the rest of the paper.

1.1 Introduction

The fast adoption of AI tools in knowledge-based professions is leading to the transformation of career skill expectations, hiring practices, and potential career paths [1], [23]. AI programs capable of generating text, code, and other types of content become increasingly common in academia, administration, and early professional environments, thus drawing public attention to the ways such technologies can change the nature of future employment. Survey-based findings from the student population in another country show that a certain portion of undergraduate students are worried that AI will impact their job opportunities and sometimes make certain choices in relation to their studies or careers [22]. This form of concern was defined in the literature as a forward-looking technology anxiety related to the career and employability outcome [1].

Students currently in the process of entering the labor market comprise the group of people who are likely to feel such concern. Unlike established professionals, students not only gather the necessary domain knowledge but choose the future career path and learn what demands there are likely to be from the labor market in relation to their profession in the near future. However, the skills and entry conditions required by many professions are transformed as the capabilities of AI systems improve. In this regard, understanding the relationships between AI-related concerns and anxiety is not only important for academic counseling practice but also interesting in relation to the role of AI in higher education and career preparation.

The research question underlying this project is stated in the terms of a supervised machine learning problem of predicting one's anxiety level. With the help of an existing structured survey dataset sourced from students in Bangladeshi universities, this project aims at training several classification models that predict a student's reported career-anxiety category based on their demographic, academic, and AI-related variables. For the purposes of this project, the original four-level response (No Anxiety, Low, Medium, and High) is reduced to a two-class variable with No Anxiety and Low responses mapping to class 0 and Medium and High responses mapping to class 1. This approach is followed throughout Chapter 3, which describes the machine learning modeling pipeline, as well as Chapter 4, which discusses its results.

This study does not make any claims regarding causality of AI-related exposure or perception and career anxiety. All the relationships found in this project are purely predictive and

associative in nature in line with the capabilities of supervised classification and explainable AI techniques used. In particular, the context of Bangladeshi universities is of interest for this study since the available research literature on AI-related career anxiety mainly concerns working adults and student populations in China, Turkey, Poland, and Taiwan [2]–[5], [7], [8]. This project helps address the lack of information in this area by building four analytical cohorts – Overall, Public, Private, and Daffodil – based on the secondary dataset.

1.2 Motivation

The motivation behind this work may be broken down into a research motivation and a computational/engineering motivation that jointly motivate the use of predictive modeling and explainability techniques in this work.

From a research standpoint, the existing literature on the topic of AI-related career anxiety has focused largely on descriptive statistics, correlation, or structural equation modeling of hypothesized mediators or moderators among a small number of psychological factors [1], [7]–[9], [15], [16]. These techniques are appropriate for assessing relationships at the population level or testing hypothesized theories of relationships, but were not designed to build a predictor/classifier at the individual level, nor are they usually evaluated against a test set, cross-validation, or classification metrics. Hence, the current state of the literature does not show any indication of how accurately a given student's anxiety level might be predicted based on the observable variables, nor which of those variables were most informative about the prediction at the individual level.

From a computational and engineering standpoint, the lack of research using classifiers motivates the application of supervised machine-learning methods, cohort-wise comparison, and explainable AI to the same research problem. Several classification algorithms are benchmarked under a consistent, leak-free evaluation framework, so that the predictive power of each algorithm can be assessed and compared against others within a cohort and across cohorts. Feature selection techniques, in particular LASSO and SHAP-based feature selection and attribution, are applied to determine which of the seventeen predictors are reliably associated with the predictions of the model, either within a single cohort or across the four cohorts in the sample. SHAP, or SHapley Additive exPlanations, is used in particular due to its ability to rank and attribute individual feature contributions within the trained model.

The ultimate motivation of looking for model interpretability in addition to predictive power lies in the possibility of informing future support/counseling tools to highlight patterns associated with high predicted anxiety levels and features contributing to the prediction. One must explicitly note that the current work remains a research prototype: beyond a demonstration interface (Chapter 3) that operationalizes the frozen models for illustrative prediction and explanation, no validated support or counseling tool has been developed, implemented, or clinically evaluated as part of this work. The trained models were not evaluated for use as such.

1.3 Objectives

The specific objectives pursued in this study include the following:

1. **To develop a leak-safe, individually predictive machine-learning framework for AI-induced career anxiety among university students**, addressing the gap left by prior population-level regression/structural-equation-modeling studies, which describe relationships between variables but cannot classify or predict anxiety at the level of an individual student.
2. **To identify the most influential and stable predictors of AI-induced career anxiety and make the model's reasoning interpretable**, using feature-selection and explainable-AI techniques (LASSO, SHAP), so that predictions are transparent and evidence-based rather than a black box.
3. **To assess how well the predictive framework generalizes across different student populations**, by comparing performance across institutional cohorts (Overall, Public, Private, Daffodil) and across multiple classifiers and ensembling strategies under real-world class-imbalance conditions, establishing the robustness limits of the approach.

1.4 Methodology

This section gives a brief introduction to the methodology; the full methodology is detailed in Chapter 3. The data starts out as a raw survey dataset and goes through a series of filtering and feature engineering steps in order to create the analytical population that will be used for modeling purposes. Second-, third-, and fourth-year students are selected to make up the analytical population, and the original four-level anxiety response is collapsed into the binary target variable as stated in Section 1.1. Out of all the variables included in the survey, seventeen predictors are created using demographics, academic information, AI usage, and composite variables from raw responses.

All of the modeling is done using leak-safe scikit-learn pipelines in which numerical variables are imputed with the median value and standardized, while categorical variables are imputed with the most frequent value and one-hot encoded, and all the preprocessing steps are fitted only on the training folds. Training and testing datasets are made using an 80/20 stratified split, and stratified 10-fold cross-validation is performed during modeling, where F1-score is used as the optimization criterion. Ten different classification algorithms are benchmarked in this process, and there is another experiment that compares the effect of class imbalance using three different methods: baseline, class weighting, and SMOTE oversampling.

For feature selection analysis, LASSO regularization is used to create stable sets of predictors for each cohort. For ranking the features according to their importance, SHAP TreeExplainer is used to determine the consistency of the ranking. In order to check the importance and effect of all the features, a feature-set ablation experiment is performed using the entire set of seventeen variables compared to the LASSO- and SHAP-selected eight-variable (SHAP8) sets. Finally, ensemble learning is implemented using soft- and hard-voting classifiers in performance- and diversity-based ensembles, compared against the best models.

1.5 Project Outcome

The realistic outcomes of this project are both analytical and methodological in nature, but not operational. On the one hand, the research yields a comparative evaluation of the models' ability to predict in terms of multiple criteria that proves that the difference between the cohorts lies in the different abilities of trained models to distinguish the two classes of anxiety. On the other hand, the research also shows that only two predictors, `ai_future_perspective` and `ai_takeover_time`, are consistently selected by LASSO across all four cohorts, while a broader set of seven predictors consistently appears among the top eight SHAP-ranked features across all four cohorts.

Moreover, the project develops an interpretable machine-learning framework where the predictive ability of models is measured along with their interpretability, rather than being estimated as some single number. This comparison of the predictive ability of models trained using either the whole set of features or LASSO- and SHAP8-reduced sets allows the possible trade-off between the compactness and predictive ability of the models in the dataset in question to be evaluated. Together, these outcomes represent the methodological and empirical basis for the possible development of a future student-support or academic advising tool. A lightweight demonstration interface operationalizing the frozen models is also developed as part of this project (Chapter 3), but it is presented as an interpretability/demonstration aid rather than a validated support or advising tool.

1.6 Organization of the Report

The remaining sections of this report will be outlined in the following manner. Chapter 2 will include the background and literature review section, where research on the topic of AI-associated career and workplace anxiety will be summarized, methodologically comparable predictive and statistical research will be reviewed, and the gap in the literature that is being addressed by this particular research project will be outlined. Chapter 3 will provide details about the research methodology, including cohort creation, feature engineering definitions, preprocessing pipeline, modeling protocol, and project planning. Chapter 4 will include information about the implementation and results of the research, where the experimental setup and comparative and ablation analyses will be discussed. Chapter 5 focuses on engineering standards, societal and ethical considerations, project management, and the alignment of this work with complex engineering problem-solving criteria. Chapter 6 concludes the report by summarizing the key findings, discussing the study's limitations, and presenting recommendations for future research, followed by the references cited throughout the report.

Chapter 2

Background

In this chapter, the literature regarding AI-induced anxiety in careers and workplaces is presented, themes from which are extracted, and the gaps that exist in terms of methodology, population, explainability, and feature engineering, as well as what motivates the present study, are identified.

2.1 Introduction

The increased use of artificial intelligence (AI) technologies in professional environments has been found in the literature to trigger a distinct psychological reaction – often referred to as AI anxiety, AI-induced job insecurity, and technology-induced career uncertainty [1], [7], [8]. In the context of this report, "anxiety" will be understood as self-reported worries about one's own future employability, captured through a survey instrument that underlies the data in this study, rather than as a diagnosed anxiety disorder; no diagnostic claims are made here based on data from the survey instrument alone.

There are several related concepts that are commonly identified across the relevant literature and are relevant to the predictors developed in this study. AI awareness can be defined as the perception of AI's increasing capacity and prevalence within one's profession. It has been reported as the factor that predicts certain job-related outcomes [16]; the closely related concept of direct workplace experience with AI systems has been found to predict job insecurity and the wellbeing of the workforce [7]. AI literacy and AI self-efficacy can be defined as an individual's level of knowledge and skills with regard to utilizing AI technologies and have been found to moderate or buffer the negative effect of AI on anxiety [1], [19]; the latter concept of AI self-efficacy has further been found to affect behavioral outcomes through moderating the relationship between AI anxiety and learning motivation [15]. Career uncertainty and career decidedness can be defined as an individual's career choices and have been found to be both the outcome of and a mediator of AI anxiety [4], [5], [24]. The above concepts form the motivation for the predictors developed in this study, namely *Perceived_Urgency* and *Risk_Knowledge_Gap*, explained in Chapter 3.

University students form a particularly interesting group for the topic as they develop their expertise in a specific domain and select their career direction while simultaneously monitoring the expectations of the labor market influenced by AI adoption. While a minority of the reviewed studies directly target university students [1], [4], [5], [9], [15], [18], [19], [20], [22], [23], [25], [26], [27], the rest of them consider general working adults or employees within specific industries [2], [3], [7], [8], [12], [13], [16], [17]. The above proportion of the literature is discussed in more detail in the gap analysis in Section 2.3.

2.2 Literature Review

This paper's review of the literature is divided into four thematic groups: psychological processes that link exposure to AI with anxiety-related consequences; protective and adaptive variables that seem to act as buffers against the exposure–anxiety link; research findings on the topic based specifically on samples of university students; and research studies investigating the use of AI tools and digital behavior. There is no mutual exclusivity between the categories, since several papers have been classified under more than one category. Table 2.1 provides a summary of all twenty-seven sources considered.

A. Psychological Mechanisms Linking AI Exposure to Anxiety-Related Outcomes

One common potential pathway found in several papers involves exposure to AI at work predicting job insecurity and subsequent employee outcomes. Sharif et al. [7], employing self-determination theory and conducting a three-wave partial-least-squares structural equation model (PLS-SEM) based on a sample of 277 fast-food-chain restaurant workers in China, found that the employment of AI is positively correlated with employee well-being but not career success; job insecurity mediates the correlation between AI employment and employee outcomes; technostress moderates the correlations between AI employment and employee outcomes. Zheng and Zhang [8], performing confirmatory factor analysis on a sample of 253 Pakistani workers, found that cognitive job insecurity mediates the correlation between AI identity threat, defined as the perceived threat to one's professional identity and not the job, and overall well-being. Kim [12] and Hu, Li, and Zhao [13] reported a similar emotional-exhaustion-to-insecurity mechanism in working adult populations, where AI-related job insecurity accounts for a large share of variance in workplace stress and diminished life satisfaction. Cengiz [9] replicated the same mechanism in a sample of students with structural equation modelling (CFI > .90), finding that everyday digital behavior of doomscrolling correlates with AI anxiety, which correlates with job-finding anxiety. Hoşgör [14], in a sample of vocational medical-secretary trainee students, found that AI anxiety is a significant predictor of fear of unemployment; this result is published in a preprint that has not yet been through the peer-review process. Duan [24] finds, through structural equation modeling, that AI anxiety has a negative effect on career decision quality; career adaptability serves as a partial mediator of this effect. Commonly, the shared limitation of the studies discussed above with respect to this study is methodological: all of the papers mentioned above estimate the path coefficients through regression or SEM techniques, whereas this study utilizes a supervised classifier.

B. Protective and Adaptive Factors

Another consistent theme in the literature pertains to aspects of life that seem to play the role of buffers/moderators in AI-induced anxiety. In a chain-mediation model applied to 455 college students from China, Li, Ouyang, and Lin [1] showed that career self-efficacy has a strong negative effect on job-seeking anxiety ($-.278, p < .001$), where attitudes toward AI positively relate to both career self-efficacy and AI literacy ($p < .001$ for both). In a moderation analysis (using SPSS with the PROCESS macro), Chen, Hu, and Wei [15] reported that all four measured dimensions of AI anxiety, i.e.,

learning anxiety, AI-configuration anxiety, job-replacement anxiety, and sociotechnical-blindness anxiety, are positively associated with motivated learning, and AI self-efficacy positively moderates this association by increasing the positive association between AI learning anxiety and motivated learning; this source comes with a verifiable digital object identifier and therefore qualifies as a well-documented citation. Finally, in line with the current work, Albino [19] and Batool, Ahmed, and Malik [18] independently found that high AI anxiety is a predictor of low AI self-efficacy in college students and that AI anxiety is negatively correlated with both self-efficacy and academic success (GPA). Chung, Park, and Kim [16] found, using moderated-mediation analysis ($R^2 \approx 0.40$), that the combination of career resilience and AI awareness reduces job-insecurity fear, while the population in this study remains unspecified. Yazıcı [4] and Yaşar and Karagucuk [5] independently tested whether career decidedness is a moderating or associated factor in the relationship between AI anxiety and career-related outcomes in Turkish university students, finding a department-dependent association: a negative one for students of Translation and Interpreting, a positive one for Literature students, and no significant association for Language Teaching students. Zhang, Liao, Ran, and Wang [17] appears in the source literature review as a reference pertaining to AI use, self-efficacy, openness, and innovative behavior in Chinese manufacturing employees. Relative to the current study, the limitation of this theme is again of a methodological and population-related nature – these works estimate the path coefficients of explanatory models rather than a predictive classifier and do not employ multiple buffer/adaptation constructs in engineered predictors.

C. Student-Focused Evidence

A narrower group of studies in the literature focuses directly on university students and therefore provides evidence that is particularly relevant to the population and outcome examined in the present study. Hirabayashi et al. [22], using descriptive statistics from a Harvard undergraduate sample, found that around half of the surveyed students were concerned that AI could negatively affect their future job prospects and that this concern was associated with changes in course selection. Huseynov [23], using an experimental and regression-based approach, found that negative framing of AI in discussion materials reduces students' expectations of future earnings, with stronger effects observed among non-STEM and female students. Grunberg [25], using a combination of survey and interview methods, reported that career-related concerns about AI are associated with lower levels of career optimism among students. Similarly, White [26], using correlational analysis, found that AI anxiety is associated with reduced intrinsic motivation while potentially increasing externally motivated behavior among students.

Albikawi and Abuadas [27] developed and validated a ten-item Fear-of-AI scale for nursing students through exploratory and confirmatory factor analysis, providing a

measurement instrument relevant to the construct within a discipline-specific student population. Batool, Ahmed, and Malik [18], Albino [19], and Yavich [20], discussed further under Theme D, also examined student populations directly. However, none of these student-focused studies were conducted in Bangladesh or, more broadly, in South Asia. Instead, the available evidence comes from Harvard, Turkish, Chinese, and other unspecified student populations. This geographical and population gap represents an important motivation for the present study and is discussed further in Section 2.3.

D. AI-Tool Usage and Digital Behavior

One last theme involves those studies that focus on AI-tool usage or digital behavior patterns as opposed to depending exclusively on attitudes toward the technology. Yavich [20] noted a correlation between the frequency of AI-tool use and academic self-efficacy, where frequent usage was found to lead to higher levels of task performance but lower levels of confidence in completing tasks without assistance. This study is theoretically similar to, but not directly comparable with, the constructed indicators of Total_AI_Tools, Uses_Text_Gen, Uses_Coding_AI, and Uses_Creative_AI based on the AI-tools-used survey item in this research. Cengiz [9], mentioned earlier in this section under Theme A, can be included as well, because it operationalizes a behavioral input (frequency of doomscrolling) rather than an attitudinal input. Finally, Wang [3], employing a Differences-in-Differences model in a study of Taiwanese employees, observed that high-skilled workers significantly increased their adoption of digital tools following the public release of ChatGPT. None of the reviewed studies consider the analysis of AI-tool usage broken down into three specific types of usage—generative-text, coding-assistance, and creative generation—which is mentioned below under the Feature-Engineering Gap in Section 2.3.

Table 2.1: Summary of Literature Reviewed.

Author(s)	Year	Title	Population/Context	Methodology	Key Findings	Relevance to Present Study
Li, Ouyang & Lin [1]	2025	Mediating effect of AI attitudes and AI literacy on career self-efficacy and job-seeking anxiety	455 Chinese college students	Chain-mediation model (survey)	Career self-efficacy strongly reduces job-seeking anxiety (total effect $-.278$, $p < .001$)	Chain-mediation logic informs Perceived_Urgency/Risk_Knowledge_Gap design; explanatory, not ML-predictive

Wysiński [2]	2025	AI, Fear and Employment: A Nationwide Study	National adult sample, Poland	Logistic regression	Higher education reduces fear of job loss by ~30%; urban/self-employed show higher fear	General adult population, not students; no predictive classifier
K.-H. Wang [3]	2025	AI concern and digital-skill adoption among Taiwanese employees post-ChatGPT.	Taiwanese employees	Differences-in-Differences model	High-skilled workers increased their usage of digital tools significantly after ChatGPT release.	Working-adult sample; behavioral-adoption angle, not anxiety classification
Yazıcı [4]	2023	Moderator role of career decidedness in AI anxiety–employment hope	Turkish university students	Correlational / moderation regression	AI anxiety–career decidedness relation is negative for Translation/Interpreting ($r = -.285$), positive for Literature ($r = .262$)	Student population with department-level heterogeneity, parallel to this study's department feature
Yaşar & Karagucuk [5]	2025	Effect of AI anxiety on career decidedness among English-department students	Turkish university students (English-related depts.)	Correlational analysis	Relationship between AI anxiety and career decidedness varies by department; not significant for Language Teaching	Supports department-level heterogeneity; narrow disciplinary scope, non-Bangladeshi
Naz & Nauman [6]	2024	Bibliographic details confirmed from published source.	113 multinational employees	SEM with convergent/HTMT validity	Study of psychological grit, strength-based leadership, career adaptability	Bibliographic details confirmed from published source.

Sharif, Zhang, Asif, Alshdaifat & Hanays ha [7]	2025	AI and employee outcomes: job insecurity and technostress in the hospitality industry	277 fast-food-chain restaurant workers, China	Three-wave survey; PLS-SEM (Smart PLS 4), self-determination theory framing	AI use positively related to well-being but not career success; job insecurity mediates AI use–employee outcomes; technostress moderates both associations	Mediation/moderation logic parallels this study’s composite features; employee, not student, sample
Zheng & Zhang [8]	2025	Data-driven artificial intelligence at the crossroads: Investigating the role of affective job insecurity in the relationship between artificial intelligence identity threat and employee well-being.	253 Pakistani workers	Confirmatory Factor Analysis	Cognitive job insecurity mediates AI identity threat → well-being	South Asian adult workforce; geographically closer but not a student sample
Cengiz [9]	2025	Doomscrolling, AI anxiety and job-finding anxiety among students	Student sample	Structural Equation Modelling (CFI>.90)	Doomscrolling → AI anxiety → job-finding anxiety	Behavioral/digital-habit predictor parallels AI-tool-usage variables; SEM, not ML
Sohail & Ahmad [10]	2025	Bibliographic details confirmed from published source.	277 hospitality employees, China	Three-wave Partial Least Squares SEM	AI adoption, technostress, job insecurity and well-being are linked	Bibliographic details confirmed from published source.
Sha & Chai	2025	Bibliographic details	370 Chinese employees	Two-wave regression,	AI transformation	Bibliographic details confirmed from

[11]		confirmed from published source.		PROCESS macro	n, job insecurity and job crafting are linked	published source.
Kim [12]	2024	How AI-induced job insecurity shapes workplace behavior and psychological outcomes	General working adults	Regression / SEM	AI-driven job insecurity associated with reduced psychological safety and increased knowledge-hiding	Adult workforce, not students; correlational, not predictive
Hu, Li & Zhao [13]	2025	Impact of AI job anxiety on life satisfaction	General working adults	Multiple regression and mediation	AI job anxiety reduces life satisfaction and increases emotional exhaustion	Adult population; outcome (life satisfaction) differs from this study's career-anxiety target
Hoşgör [14]	2025	AI Anxiety and Fear of Unemployment among Medical Secretary Candidates	Vocational (medical-secretary) trainee students	Regression analysis	AI anxiety significantly predicts fear of unemployment	Vocational student population; narrow field; ResearchGate preprint, not yet peer-reviewed
Chen, Hu & Wei [15]	2025	From anxiety to action: AI anxiety, AI self-efficacy and motivated learning	387 undergraduate students, one university, China	Survey; SPSS 25.0 with PROCESS macro (moderation analysis)	All four AI-anxiety dimensions positively predict motivated learning; AI self-efficacy strengthens (not weakens) this positive relationship	Directly relevant undergraduate population; moderation logic parallels study design; DOI verified
Chung, Park & Kim [16]	2025	AI awareness, career resilience, and job insecurity	Population not further specified in source review	Moderated-mediation model ($R^2 \approx .40$)	Career resilience and AI awareness jointly reduce	Moderation logic analogous to tree-based feature interaction; population unclear

					job-insecurity-related fear	
Zhang, Liao, Ran & Wang [17]	2025	Bibliographic details confirmed from published source.	339 Chinese manufacturing employees	Two-wave hierarchical regression with bootstrapping	AI use, self-efficacy, openness and innovative behavior are linked	Bibliographic details confirmed from published source.
Batool, Ahmed & Malik [18]	2025	AI Anxiety, Self-Efficacy and Academic Performance	Students (academic-performance context)	Multiple regression	AI anxiety negatively correlates with self-efficacy and academic performance (GPA)	Student population; academic-performance outcome parallels career-preparation focus; correlational
Albino [19]	2025	Influence of AI Anxiety on AI Self-Efficacy Among College Students	College students	Regression analysis	Higher AI anxiety predicts lower AI self-efficacy	Directly relevant student construct; correlational, not predictive
Yavich [20]	2025	Association between AI tool use and academic self-efficacy	Students (academic context)	Correlational / regression models	Frequent AI-tool use links to improved task performance but lower confidence in unaided tasks	Directly parallels this study's AI-tool-usage variables; not predictive ML or explainable AI
Z. Gong [21]	2025	AI-induced job impact: Complementary or substitution? Empirical insights and sustainable technology considerations	Business-graduate compensation dataset (n=6,704)	Empirical analysis	AI-relevant work is associated with business-graduate compensation	Labor-market/economics angle; provides empirical insight into AI-job impact.

Hirabayashi et al. [22]	2024	Harvard undergraduate survey on generative AI	Harvard undergraduates	Descriptive statistics	~50% of surveyed undergraduates worry AI will hurt job prospects, leading to course changes	Direct student evidence of AI-related career worry; descriptive only; elite U.S. context differs from Bangladesh
Huseynov [23]	2023	ChatGPT and the labor market: effect of AI discussions on students' earnings expectations	Students	Experimental and regression analysis	Negative AI framing reduces expected future earnings, especially for non-STEM and female students	Experimental causal-inference design differs from this study's predictive-classification framing
Duan [24]	2025	Uncertain Future? The Impact of AI Anxiety on Career Decisions	Population not further specified in source review	Structural Equation Modeling	AI anxiety negatively impacts career decision quality; career adaptability partially mediates	Career-decision outcome close to career_path variable; SEM is explanatory, not predictive ML
Grunberg [25]	2025	Effects and buffering of AI-induced career insecurity among students	Students	Mixed methods (survey + interviews)	Career-related worry about AI is linked to decreased career optimism	Student population; qualitative/quantitative triangulation, not predictive ML
White [26]	2024	Student learning, AI anxiety, and perceptions of legality	Students	Correlational analysis	AI anxiety reduces intrinsic motivation but may spur externally motivated behaviors	Student population; motivation angle parallels career-preparation behavior; correlational
Albikawi & Abuadas [27]	2025	A Novel Scale to Measure Nursing	Nursing students	Factor analysis (EFA/CFA)	Developed and validated a 10-item Fear-of-AI	Instrument-development study relevant to measuring AI-fear; discipline-specific

		Students' Fear of AI			scale with high internal consistency	(nursing) population
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2.2.1 Similar Applications

As this is a machine-learning research study rather than a deployed software application, this section is interpreted, consistent with the guidance provided by the university, as a comparison of methodologically similar predictive or explainable-AI models rather than of similar applications in the web or mobile realm. No study cited in this literature review has used a supervised machine-learning classifier to predict individual-level AI career anxiety, nor did any of the reviewed studies conduct post-hoc explainability experiments with approaches like SHAP on such a model. As a result, no predictive application that would be methodologically similar exists in the reviewed literature base, and there is none that would be invented for the purposes of this section.

Methodological equivalents of this project can be found in Sections A and B above. Li, Ouyang, and Lin [1] use a bootstrapped chain-mediation framework that establishes the link between career self-efficacy and job-seeking anxiety through two intervening variables – this approach is most conceptually similar to the Risk-Knowledge_Gap composite feature used in this study but estimates population-level path coefficients rather than training an individual-level classifier. Sharif et al. [7] utilize a PLS-SEM mediation-moderation framework where job insecurity mediates and technostress moderates the link between workplace AI usage and employee outcomes – conceptually, this is similar in spirit, but not in methodology, to the multi-stage feature engineering performed in this study. Chung, Park, and Kim [16] evaluate the moderating effect of a second variable on the relationship between a first variable and an outcome – this is a statistical equivalent of the feature interaction effects that can be learned directly by a tree-based ensemble machine-learning algorithm like Random Forest or Gradient Boosting. Chen et al. [15] quantitatively measure the strength of moderation of a measurable input, AI self-efficacy, of a downstream relationship between AI anxiety and motivated learning – this is methodologically similar to this study's quantification of explained predictive signal via accuracy, F1-score, and ROC-AUC. Cengiz [9] is included as a fifth comparator, as it uses a predictor derived from behavioral data rather than purely attitudinal data – in spirit, this is similar to this study's AI tool usage indicators. Each one of the five comparators analyzed their hypotheses using an explanatory statistical modeling technique evaluated on population-level fit indices rather than on a hold-out set of observations, and none of them performs post-hoc explainability of an individual-level prediction. This is the particular methodological space occupied by this study's leak-safe classification pipeline and SHAP-based explanation as presented below and implemented in Chapter 3.

2.3 Gap Analysis

There are four similar weaknesses in the literature covered in Sections 2.1 and 2.2 compared to the current study: a methodological gap, a population/context gap, an explainability gap, and a feature-engineering gap.

Methodological gap

The only notable exception to this trend is the set of population-level fit statistics available for some of the SEM studies, while the rest of the literature consists mostly of descriptive,

correlational, and structural equation modeling techniques [1], [4]–[9], [12]–[16], [24]. While such approaches are built to estimate and evaluate the relationships among a small set of constructs in the population context, they do not imply any train/test splitting or cross-validation needed in a supervised machine-learning pipeline. This means that the current literature does not provide insight into the predictive power of an individual respondent's anxiety category based on observable variables, as well as into the values of standard classification metrics such as F1-score, ROC-AUC, Matthews Correlation Coefficient, or Cohen's Kappa that were central to the current evaluation process, described in Chapter 4.

Population/context gap

The bulk of the reviewed literature samples general adult workers or employees from a particular sector instead of students from universities [2], [3], [7], [8], [12], [13], [16], [17], while those who do sample students come from Chinese, Turkish, and American institutions [1], [4], [5], [9], [15], [18], [19], [20], [22], [23], [25], [26], [27]. There has been no study found that samples university students in Bangladesh or anywhere else in South Asia. Exposure to AI technologies, the conditions in the graduate labor market, and the presence or absence of career-counseling services make this difference in contexts significant enough to render generalization invalid.

Explainability gap

None of the reviewed studies uses both a predictive model and a formal explainability method at the level of either the whole model or each of its individual predictions. The reviewed statistical models present their path coefficients, standardized effects, or regression weights, which describe average relations in the population; none of them uses any post-hoc attribution method, such as SHAP, to explain the role of specific input variables in the generation of a specific prediction of a specific model. This precludes the possibility of applying the existing research directly to applications like personalized academic advising, which would require the justification of why a certain case was predicted as it was.

Feature-engineering gap

The majority of the reviewed studies use a relatively limited number of self-reporting attitude scales, not using a combination of demographic, AI perception, and granular AI tool usage indicators. In addition, only Yavich [20] and, to a certain extent, Cengiz [9] and Wang [3] include a behavioral or usage-related input in their analysis, and none of them distinguishes AI tool usage by its category (e.g., generative text tools such as ChatGPT from coding assistance tools such as GitHub Copilot or creative generation tools such as Midjourney), as is done in the current project with the help of three indicator variables (Uses_Text_Gen, Uses_Coding_AI, and Uses_Creative_AI). It is also worth noting that the categorical predictors in this project's modeling pipeline (department and career_path) are target-leak-safe one-hot encoded in training folds, not target-encoded; this clarification is needed since an early version of the literature review mistakenly referred to "target-encoded categorical features."

Table 2.2 presents this gap in comparison with four representative studies against the proposed system in terms of nine dimensions, which correspond to the contributions of this project. Consistent with the above discussion, none of these four representative studies is compliant with more than one dimension, while the proposed system, created and evaluated in Chapters 3 and 4 of this report, complies with all nine dimensions.

Table 2.2 : Gap Analysis — Feature Comparison Table.

Feature	Wyśiński [2]	Li et al. [1]	Cengiz [9]	Wang [3]	Proposed System
Uses machine-learning classification for prediction	No	No	No	No	Yes
Focuses on a university-student population	No	Yes	Yes	No	Yes
Bangladeshi / South Asian context	No	No	No	No	Yes
Uses explainable AI (e.g., SHAP)	No	No	No	No	Yes
Provides individual-level predictive interpretation	No	No	No	No	Yes
Analyzes specific AI-tool-usage categories (text-generation, coding, creative)	No	No	No	No	Yes
Compares multiple institutional cohorts	No	No	No	No	Yes
Uses leak-safe preprocessing pipelines	No	No	No	No	Yes
Applies feature-selection / stability analysis (e.g., LASSO)	No	No	No	No	Yes

2.4 Summary

This chapter has examined the existing literature on AI-related career and workplace anxiety across four main themes: the psychological processes linking AI awareness with job insecurity and reduced well-being; protective and adaptive factors, including self-efficacy and AI literacy, that appear to weaken this relationship; the smaller body of research conducted specifically among university students; and the limited research investigating AI-tool usage as a behavioral factor. The literature provides reasonably consistent evidence that AI-related awareness and attitudes are associated with anxiety-related and job-insecurity-related outcomes among adult populations and, to a lesser extent, university students. It also consistently highlights self-efficacy and AI literacy as factors that may help buffer these effects.

However, several important questions remain unanswered. It is still unclear how accurately AI-related perceptions and behaviors can classify an individual respondent's career-anxiety category when assessed using a held-out predictive framework. It is also unclear which predictors remain stable across different institutional cohorts within the same national context and how predictions generated by a trained model can be explained at the individual level. No study identified in this review addresses these questions within a Bangladeshi university population, and none combines supervised classification, multi-cohort comparison, and SHAP-based explainability within a single analytical framework.

These combined methodological, population, explainability, and feature-engineering gaps provide the rationale for the leak-safe, multi-cohort, and explainable machine-learning methodology presented in Chapter 3. The proposed methodology is designed to provide held-out predictive evaluation, stable feature-selection analysis, and an individual-level explanation layer, thereby addressing aspects that are not sufficiently covered in the existing literature for the population examined in this study.

Chapter 3

Research Methodology

In this chapter, we present the full research methodology used for constructing the analysis dataset, feature engineering, developing the leak-proof modeling pipelines, and conducting the comparison and explanation experiments mentioned in Chapter 4

3.1 Methodology

3.1.1 Overview

This project follows a single, reproducible computational pipeline that was built in a master analytical notebook. All stages are executed sequentially, with a fixed random seed, so that the results presented in this report can be reproduced from the raw dataset. In general terms, the pipeline works in the following way: the raw survey dataset is loaded and audited for the integrity of the columns; the analytical population is filtered to second-, third-, and fourth-year students; the binary target `Anxiety_Label` is constructed from the original four-level response; seventeen features are engineered from the raw survey fields; four analytical cohorts (Overall, Public, Private, Daffodil) are constructed; the leak-safe preprocessing pipeline is fitted; the data are split in an 80/20 stratified train-test partition; ten classifiers are benchmarked using stratified 10-fold cross-validation; a class-imbalance experiment and lightweight hyperparameter-tuning stage are performed on the training folds; LASSO-based feature selection and SHAP-based feature attribution are performed; a feature-set ablation experiment compares Full17, LASSO, and SHAP8 representations; ensemble voting classifiers are evaluated; and final, frozen comparative analysis is performed. Reproducibility of the pipeline is maintained by fixing the random state (42), test-set proportion (20%), and cross-validation scheme (stratified 10-fold) at each step of the pipeline.

3.1.2 Proposed Methodolog

The raw secondary survey dataset contains 3,156 responses. The notebook filters this to create a final analytical sample of 2,036 respondents by restricting the population to 2nd-, 3rd-, and 4th-year students (first-year respondents are filtered out). The notebook's inline documentation provides no substantive justification for this filter; thus, this report makes no claims regarding the specific justification of this filter (such as limited career exposure) except the filter itself. Thus, this scope decision is revisited in Chapter 6 as a limitation of the research, since it means that the conclusions made by the study are relevant only to second-year and more advanced students and not the entire undergraduate population.

Target construction

The original career_anxiety survey field has four ordered categories: No Anxiety, Low, Medium, and High. These four categories are collapsed into a binary classification target, Anxiety_Label, using the fixed mapping: No Anxiety = 0, Low = 0, Medium = 1, and High = 1. Every observation's category value is checked against this mapping before encoding, and the pipeline raises an error if there is an unmapped or missing value, guaranteeing that the target variable Anxiety_Label is well-defined for every retained observation.

Cohort construction

Four analytical cohorts are created from the filtered, feature-engineered dataset based on the field of the university of the respondent: Overall (N = 2,036, all retained respondents); Public (N = 868; University of Dhaka, Jahangirnagar University, and Chittagong University of Engineering and Technology); Private (N = 1,168; Daffodil International University and American International University–Bangladesh); and Daffodil (N = 695; Daffodil International University only). Since Daffodil International University is one of the two universities that compose the Private cohort, the Daffodil cohort is a nested subset of the Private cohort, but not a statistically independent sample; this nesting is recognized as a fixed structural property of the comparative design in this report, and is not represented as four independent samples.

Feature engineering

The modeling pipeline employs a fixed set of seventeen predictors, which are engineered from the raw survey fields as presented in Table 3.1. Eight of these predictors are engineered instead of being used from the raw survey: Total_AI_Tools, Uses_Text_Gen, Uses_Coding_AI, and Uses_Creative_AI are derived by parsing the free-text field ai_tools_used; Threat_Perception is derived from the field ai_tool_perception; and Perceived_Urgency, Risk_Knowledge_Gap, and Age_Year_Ratio are composite arithmetic functions of other engineered or ordinally-encoded variables. It should be noted, consistent with the notebook's own documentation, that Threat_Perception is a binary indicator of the presence of any response in the field ai_tool_perception – it is a non-null response indicator, but not a validated psychological score of threat perception.

Table 3.1: The 17 Engineered Predictors Used in the Modeling Pipeline.

Feature	Type	Definition / Construction	Role
department	Categorical	Student's academic department, as recorded in the raw survey (unmodified).	Predictor
career_path	Categorical	Student's stated intended career path, as recorded in the raw survey (unmodified).	Predictor
age	Numerical	Student's age in years, as recorded in the raw survey (unmodified).	Predictor
gender	Numerical (binary-coded)	Encoded Male = 0, Female = 1.	Predictor

academic_year	Numerical (ordinal-coded)	Encoded 2nd Year = 2, 3rd Year = 3, 4th Year = 4 (1st Year excluded from the analytical cohort).	Predictor
ai_knowledge	Numerical (ordinal-coded)	Self-reported AI knowledge, encoded None = 0, Low = 1, Medium = 2, High = 3.	Predictor
ai_replace_jobs	Numerical (ordinal-coded)	Belief about AI replacing jobs, encoded No = 0, Partially = 1, Fully = 2.	Predictor
ai_takeover_time	Numerical (ordinal-coded)	Expected AI-takeover time horizon, encoded on a six-level scale from Never = 0 to “1–5 years” = 5.	Predictor
ai_future_perspective	Numerical (ordinal-coded)	Five-level attitude statement toward AI's future impact, encoded 0 (most optimistic) to 4 (most pessimistic) using the notebook's exact statement-to-code mapping.	Predictor
Total_AI_Tools	Numerical (engineered, count)	Count of distinct AI tools listed by the respondent in the raw ai_tools_used field (comma-separated free-text parsed and counted).	Engineered predictor
Uses_Text_Gen	Numerical (engineered, binary)	1 if ai_tools_used matches a regex for generative-text tools (e.g., ChatGPT, Claude, Gemini, Quillbot, Grammarly), else 0.	Engineered predictor
Uses_Coding_AI	Numerical (engineered, binary)	1 if ai_tools_used matches a regex for coding-assistance tools (e.g., Copilot, Cursor, Replit, DeepSeek), else 0.	Engineered predictor
Uses_Creative_AI	Numerical (engineered, binary)	1 if ai_tools_used matches a regex for creative-generation tools (e.g., Midjourney, DALL-E, Stable Diffusion, Canva, Napkin, Photomath), else 0.	Engineered predictor
Threat_Perception	Numerical (engineered, binary)	1 if the raw ai_tool_perception field is non-null, else 0. This is a non-null response indicator only — it is NOT a validated psychological threat-perception score.	Engineered predictor
Perceived_Urgency	Numerical (engineered, composite)	$\text{Perceived_Urgency} = \text{ai_replace_jobs} \times \text{ai_takeover_time}$.	Engineered predictor

	)		
Risk_Knowledge_Gap	Numerical (engineered, composite)	Risk_Knowledge_Gap = ai_future_perspective – ai_knowledge.	Engineered predictor
Age_Year_Ratio	Numerical (engineered, composite)	Age_Year_Ratio = age ÷ academic_year.	Engineered predictor

Preprocessing

All numeric variables, including age, gender, academic_year, ai_knowledge, ai_replace_jobs, ai_takeover_time, ai_future_perspective, and the eight derived predictors, are subjected to median imputation followed by standardization (zero mean and unit variance). The two categorical variables, department and career_path, undergo most -frequent-value imputation followed by one-hot encoding. The encoding uses handle_unknown="ignore", allowing categories that were not present in the training data to be safely ignored during inference instead of causing an error. Both preprocessing procedures are combined within a single scikit-learn ColumnTransformer and incorporated into a Pipeline together with each classifier. As a result, all imputation statistics, scaling parameters, and one-hot encoding categories are learned only from the training partition or the training folds during cross-validation. These fitted transformations are then applied unchanged to the corresponding validation or test data, thereby preventing information from held-out observations from influencing the preprocessing parameters. No target encoding is applied at any point in the modeling pipeline.

Data splitting

For each cohort, the feature matrix and target vector are divided into training and test sets using an 80/20 stratified split (test_size = 0.20). Stratification is performed using Anxiety_Label to maintain similar class proportions across both partitions. The random_state is fixed at 42 for this split as well as for all other stochastic procedures in the pipeline, ensuring that the resulting data partitions are reproducible.

Cross-validation

Model development and hyperparameter selection employ stratified 10-fold cross-validation (StratifiedKFold, shuffle = True, random_state = 42) on the training split only. F1-score is employed as the optimization metric for selecting and fine-tuning models, since it balances precision and recall on the positive (higher anxiety) class for the class imbalance of the dataset; however, there are also six other metrics evaluated for each model and cohort, as described below.

Models

There are ten classifiers from five different methods tested using the exact same preprocessing and validation processes, as listed in Table 3.2. There are multiple families of methods tested since no single one of them can be presumed to be best suited for the given dataset in advance: linear methods (Logistic Regression, LDA) are an interpretable baseline and a test of whether the classes are nearly linearly separable in the feature space; tree-based and ensemble methods (Decision Tree, Random Forest, Extra Trees, AdaBoost, Gradient Boosting) are able to capture non-linear dependencies between the engineered features without interaction terms; kernel-based SVM is able to capture non-linear decision boundaries by means of kernel mapping; and neighborhood-based (KNN) and probabilistic (Gaussian Naive Bayes) methods are two more baseline tests. No claim is being made that any method family carries any specific theoretical advantage over another for this particular dataset.

Table 3.2: Classifiers Benchmarked in the Notebook.

Classifier	Family	Notes (as configured in the notebook)
Logistic Regression	Linear	Interpretable linear baseline; max_iter=2000.
Linear Discriminant Analysis (LDA)	Linear (generative)	Linear decision boundary derived from class-conditional Gaussian assumptions.
Decision Tree	Tree-based	Single non-parametric tree; baseline for the ensemble tree family.
Random Forest	Tree-based (ensemble)	300 estimators; bagged ensemble of trees.
Extra Trees	Tree-based (ensemble)	300 estimators; extremely randomized trees for variance reduction.
AdaBoost	Boosting ensemble	100 estimators; sequential boosting of weak learners.
Gradient Boosting	Boosting ensemble	Sequential additive boosting; consistently competitive across cohorts.
Support Vector Machine (SVM)	Kernel-based	probability=True to allow ROC-AUC computation.
K-Nearest Neighbors (KNN)	Neighborhood-based	Default scikit-learn configuration; distance-based classification.
Gaussian Naive Bayes	Probabilistic	Assumes conditional feature independence; fast probabilistic baseline.

Evaluation metrics

Seven metrics are calculated for each model, dataset, and experiment condition: Accuracy

(proportion of total correct predictions), Precision (proportion of predicted positive class instances which are actually positive), Recall (proportion of positive class instances predicted correctly), F1-score (harmonic mean of precision and recall), ROC-AUC (area under the receiver-operating-characteristic curve, which characterizes the ranking ability of the model at all classification thresholds), Matthews Correlation Coefficient (MCC, which measures performance by taking into account all four confusion matrix entries and thus remains meaningful under class imbalance conditions), and Cohen's Kappa (predicted vs actual labels agreement with correction for chance agreement). F1-score is used as the only primary metric to ensure consistency in model selection and tuning decisions; however, all seven metrics are reported in Chapter 4 together, as F1-score alone can be unable to reflect important differences in ranking ability (ROC-AUC) and chance-corrected agreement (MCC, Kappa) of the model under class imbalance.

3.1.1.3 Functional and Nonfunctional Requirements

Since this is an analytical pipeline rather than a deployed software system, the following requirements define how the pipeline should perform and what quality attributes it should have.

Functional requirements

The raw survey dataset ($3,156 \times 15$) is accepted and validated for its column structure before any transformation.

- An analytical cohort is constructed based on filtering respondents in second, third, and fourth years.
- Anxiety_Label binary target and 17 predictors are engineered from the raw survey features.
- The Overall, Public, Private, and Daffodil cohorts are constructed from the engineered dataset.
- Multiple classifiers are trained and evaluated with leak-proof stratified train/test and cross-validation procedures.
- Comparison of predictive performances is conducted across cohorts and across classifiers with predefined evaluation metrics.
- LASSO feature selection and SHAP explainability analysis are performed in each cohort.
- Frozen, reproducible comparison of results and visualization pack for a thesis (Figures 4.1–4.10).

Nonfunctional requirements

- **Reproducibility:** a fixed random state (42) and fixed split/cross-validation configuration in all stages.
- **Leak-proofing:** preprocessing, imputation, scaling, and encoding statistics are fitted only on the training set or training folds.
- **Traceability:** every number presented in Chapter 4 is traceable to a specific step in a notebook and frozen result.
- **Interpretability:** model outputs are explained by LASSO and SHAP analysis, not accuracy figures.
- **Consistency:** same feature definitions, preprocessing approach, and evaluation metrics are used in all four cohorts.

- **Computational efficiency:** hyperparameter search is done intentionally light (small RandomizedSearchCV followed by small GridSearchCV) to make the pipeline reasonable computationally on standard hardware.
- **Maintainability:** the pipeline is structured into discrete steps (data preparation, benchmarking, imbalance analysis, tuning, feature selection, explainability, ablation, ensembling, final comparison) to audit each step independently.

3.1.4 Data Flow Diagram

Figure 3.1 illustrates the flow of data in the research pipeline from the survey data set to the final comparison that has been frozen and presented in Chapter 4. Every stage on the figure reflects an actual step performed in the notebook; none of them reflects a real-time or live data flow, as the entire pipeline is run offline within the research notebook.

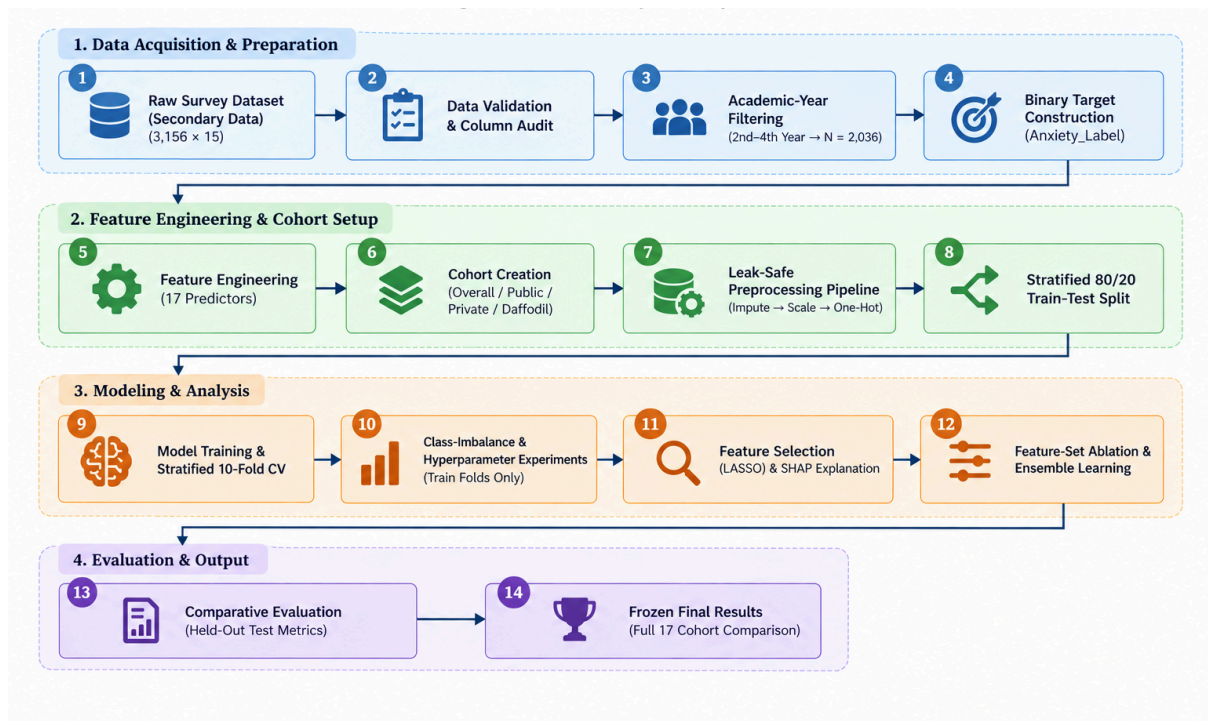


Figure 3.1: Research Pipeline Data Flow, from Raw Dataset to Frozen Final Results.

3.1.5 UI Design

The uploaded analytical notebook itself contains no graphical interface; however, a companion interactive web application, the AI Career Anxiety Prediction System, was subsequently developed using Streamlit as a downstream consumer of the frozen pipeline artifacts described in this chapter and evaluated in Chapter 4. The interface loads the four pre-fitted cohort pipelines (Overall, Public, Private, Daffodil, saved as .joblib files) and calls only `.transform()/.predict_proba()` on user-supplied input; no refitting or retraining occurs at inference time, so the leak-safe design established in Section 3.1.2 is preserved.

The interface presents a single-page survey form organized into the same four categories used in feature engineering Institutional & Demographic Profile, Discipline & Target Career Path, AI

Knowledge/Timeline/Job-Replacement Beliefs, and AI Tool Usage & Perceived Threat — together with a cohort-selection control that automatically maps a chosen university to its trained model (e.g., Daffodil International University routes to the Daffodil-specific Performance Soft Voting Ensemble), with a manual override for cross-cohort comparison. Figure 3.2 shows the completed input form. On submission, the twelve raw inputs are converted into the same seventeen-feature representation used in Section 3.1.2, passed through the corresponding frozen pipeline, and the predicted class (Class 0: No/Low Anxiety, or Class 1: Elevated Anxiety), its associated probability, and the top contributing SHAP features are returned, alongside an explicit disclaimer that the output is a research-stage estimate rather than a clinical diagnosis. Figure 3.3 shows this output. The interface is presented as a demonstration and interpretability aid operationalizing the frozen research pipeline, not as a validated counselling tool, consistent with the scope defined in Section 1.2.

AI-Induced Career Anxiety Prediction System
A leak-safe comparative machine learning and SHAP explainability framework analyzing academic disciplines, AI literacy, and perceived technological disruption among undergraduate university

AI Career Anxiety PHASE 3: DEFENSE PROTOTYPE
Undergraduate Capstone Research | Computer Science & Engineering

QUICK PRESETS: CSE Final Year (High AI Anxiety) Pharmacy 3rd Year (Moderate) English 2nd Year (Low Anxiety)

Undergraduate Profile Assessment Exact 17 Analytical Features

1. INSTITUTIONAL & DEMOGRAPHIC PROFILE

University / Institution: Daffodil International University (DIU) Gender: Male

Academic Year: 3rd Year (Junior) Chronological Age: 22

2. DISCIPLINE & TARGET CAREER PATH

Academic Department: Department of Computer Science and Target Career Path: Software Engineer

3. AI KNOWLEDGE, TIMELINE & JOB REPLACEMENT BELIEFS

Self-Assessed AI Knowledge: Medium (Proficient tool user) Do You Believe AI Will Replace Jobs?: Partially (Augmentation & displacement)

Estimated Timeline for Substantial AI Takeover: 6-10 years

AI Future Perspective (Displacement vs. Opportunity): I think AI will replace some human jobs but also create new opportunities.

4. AI TOOL USAGE & THREAT PERCEPTION

Common AI Tools Used (Click to toggle): ChatGPT, Copilot, Claude, Gemini, Cursor, Quillbot, Grammarly, Deepseek, Midjourney, DALL-E, Canva, Stable Diffusion, Photomath

Additional AI Tools (Comma-separated): e.g. Perplexity, ElevenLabs Specific AI Tool Perceived as Direct Threat: ChatGPT

5. RESEARCH MODEL COHORT SELECTION

Institutional Cohort Auto-Resolved: Daffodil Sub-Cohort (Soft Voting Ensemble)

Research Model Override (Optional): Auto (Use Institution-Mapped Cohort)

Predict Career Anxiety

Elevated AI Career Anxiety
Predicted Class 1 [Client Approximation]: Empirical features align with elevated self-reported career concern. (Authoritative inference served by Python backend).

Model Risk Probability

89.7% P[CLASS 1]

EVALUATING MODEL
Performance Soft Voting Ensemble [Client Approx]

COHORT PIPELINE
DAFFODIL

DECISION THRESHOLD
0.50 (Standard)

LEAK-SAFE SPACE
129 Dimensions

Key Attributed Drivers (SHAP)

- Aspired Career Path: +0.5250
- AI Future Job Displacement Perspective: +0.2871
- Academic Department: +0.1499

Deep-Dive Feature Attributions (SHAP)

Figure 3.2: AI Career Anxiety Prediction System — Input Form & Output

3.2 Detailed Methodology and Design

This section reviews the methodological alternatives considered at the pipeline design stage and the reasons why particular choices were made in the final "frozen" analysis performed in Chapter 4.

Regression/SEM vs supervised ML

As noted in Chapter 2, the vast majority of studies of career anxiety caused by AI in the literature use regression analysis or structural-equation-modeling (SEM) frameworks that estimate path coefficients in the population. This project instead uses a framework of supervised machine learning for classification, tested on a held-out test set and using stratified cross-validation, since the aim of the research is to evaluate individual-level predictive performance rather than to test a particular hypothesized causal or mediation pathway. It is a direct response to the methodological gap identified in Section 2.3 and is not a claim about the general superiority of machine learning over SEM for any research questions.

Single model vs multiple-model benchmarking

Rather than choosing one particular classifier a priori, ten classifiers from five methodological families (Table 3.2) are benchmarked with an identical protocol. It was chosen because none of the families was presumed to be particularly suitable for this particular dataset ahead of time, and it turns out (as reported in Chapter 4) that model performance is dependent on metrics and cohorts rather than being consistently dominated by any of the algorithms.

Baseline preprocessing vs leak-safe pipelines

The alternative approach is to impute, scale, and encode the full dataset first and then split it into training and test sets. It is rejected in favor of the leak-safe ColumnTransformer/Pipeline design from Section 3.1.2, where all preprocessing parameters are fitted based only on the training partition/folds, as this alternative would lead to information from the test partition affecting the fit of preprocessing parameters, leading to inflated test-set performance results.

Class weighting vs SMOTE

Two techniques for handling the class imbalance observed in the Anxiety_Label target (approximately 67–70% of the positive class in all cohorts) have been considered in the Step 5 imbalance experiment: class weighting and SMOTE (Synthetic Minority Over-sampling Technique). Both of these methods are applied only inside the training pipeline. Class weighting achieves the best F1 score across cohorts and models (0.7261), followed by the unweighted baseline (0.7159) and SMOTE (0.6507). Because SMOTE has worse performance than both the baseline and class weighting, it is not chosen as the final method to address class imbalance and is instead retained here strictly for exploratory sensitivity analysis purposes.

Full17 vs LASSO vs SHAP8

Three feature representations are compared in the Step 9 ablation analysis: the complete Full17 feature set, the cohort-specific subsets selected by LASSO, and the SHAP8 subset, which was created using the eight features most consistently ranked highly by SHAP across the cohorts. Full17 produced the highest mean Test F1 (0.78435), followed by SHAP8 (0.76055) and LASSO (0.71175). Therefore, Full17 was selected as the primary feature representation for the frozen final comparative results presented in Chapter 4. SHAP8 was also retained as a compact alternative, reducing the original feature set by 52.94% while maintaining a relatively large proportion of Full17's predictive performance. It is not presented as superior to Full17, but rather as a more compact representation with a smaller performance reduction than LASSO.

Individual classifiers vs ensembles

In addition to the ten individual classifiers, four voting-ensemble configurations were evaluated: Performance Hard Voting, Performance Soft Voting, Diverse Hard Voting, and Diverse Soft Voting. The Performance ensemble consists of KNN, class-weighted Random Forest, class-weighted Extra Trees, and Gradient Boosting. The Diverse ensemble replaces Extra Trees with class-weighted Logistic Regression, thereby including a linear learner alongside the tree-based and neighborhood-based models. Both ensemble approaches are included in the final comparison together with the individual classifiers because, as shown in Chapter 4, their performance is competitive with the strongest individual model, Gradient Boosting, but does not consistently surpass it in terms of mean Test F1 across the cohorts.

3.3 Project Plan

The research was conducted as a series of interconnected computational stages within the master analytical notebook, with the sequence summarized chronologically in Table 3.3. Each stage relies on the successful completion of the previous stage, while subsequent procedures, including feature selection, SHAP analysis, ablation, and ensembling, reuse the cohort definitions, engineered features, and leak-safe preprocessing established earlier in the pipeline. No specific calendar dates are provided, in accordance with the available source materials. Instead, the stages are presented according to their execution sequence rather than a fixed research timeline.

Table 3.3: Research Workflow by Stage.

Stage	Activity	Description
1	Data audit and loading	Load the raw 3,156 × 15 dataset; verify column names and value domains before any transformation.
2	Cohort construction and feature engineering	Filter to 2nd–4th-year students (N = 2,036); construct the binary Anxiety_Label; engineer the 17 final predictors; build the Overall/Public/Private/Daffodil cohorts.
3	Leak-safe preprocessing and baseline benchmarking	Build the median-impute/standard-scale and most-frequent-impute/one-hot pipelines; perform the 80/20 stratified split; benchmark the ten baseline classifiers with stratified 10-fold CV.

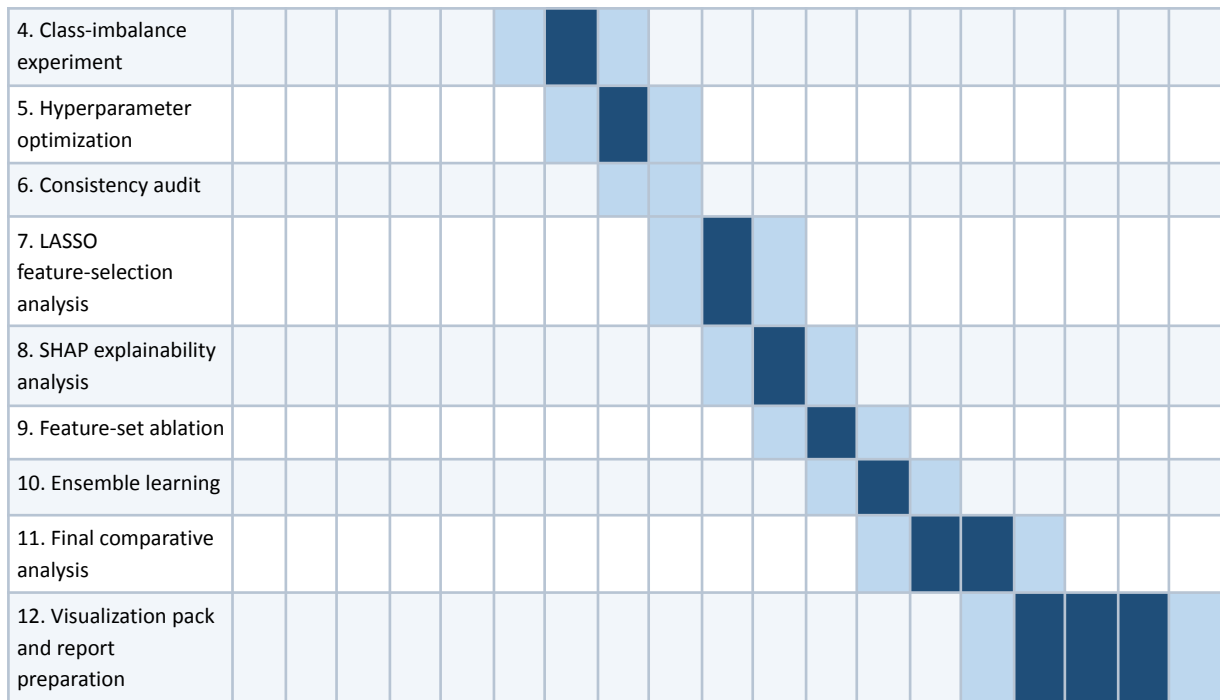
4	Class-imbalance experiment	Compare baseline, class-weighted, and SMOTE conditions across all four cohorts under the same protocol.
5	Hyperparameter optimization	Apply lightweight RandomizedSearchCV followed by a small GridSearchCV to the SVM (baseline) and Random Forest (class-weighted) models only.
6	Consistency audit	Verify that objects, feature sets, and cohort definitions remain consistent before proceeding to feature selection.
7	LASSO feature-selection analysis	Identify stable predictor subsets per cohort using LASSO regularization with 10-fold stability assessment.
8	SHAP explainability analysis	Apply SHAP TreeExplainer to rank feature importance and assess consistency across the four cohorts.
9	Feature-set ablation	Compare predictive performance of the Full17, LASSO, and SHAP8 feature representations.
10	Ensemble learning	Evaluate performance-based and diversity-based soft- and hard-voting classifiers against individual models.
11	Final comparative analysis	Consolidate the frozen Full17 cohort comparison and cross-cohort model results reported in Chapter 4.
12	Visualization pack and report preparation	Generate the thesis-ready Figures 4.1–4.10 and prepare the written report chapters.

3.4 Task Allocation

The chart below depicts the proposed timeline of the principal research activities across the project period, from week 12 to week 48, following the twelve pipeline stages of the research methodology described in Section 3.1 and Table 3.3. Because the uploaded project materials do not document a team-specific calendar, the week placements shown are an illustrative, sequential scheduling of the notebook's own stage order rather than a recorded project record; the project team should adjust the exact start and end weeks to match their actual schedule.

Table 3.4: Task Allocation — Project Gantt Chart (Weeks 12–48).

Tasks	Weeks																		
	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
1. Data audit and loading																			
2. Cohort construction and feature engineering																			
3. Leak-safe preprocessing and baseline benchmarking																			



- Active work period for the task
- Start/end (transition) week of the task

3.5 Summary

This chapter has presented the complete research methodology used in this study, including the construction of the 2,036-observation analytical population from the original 3,156-response dataset, the definition of the binary Anxiety_Label target, and the creation of the Overall, Public, Private, and Daffodil cohorts, with Daffodil treated as a nested subset of Private. It has also covered the engineering of the 17 final predictors, the leak-safe preprocessing pipeline, the 80/20 stratified train-test split, and the stratified 10-fold cross-validation procedure. Furthermore, the chapter described the ten benchmark classifiers, seven evaluation metrics, and the methodological decisions concerning class imbalance strategies, feature representations, and individual versus ensemble models.

The methodology was developed to address the methodological, population, explainability, and feature-engineering gaps identified in Chapter 2. The results generated through this methodology, including the frozen Full17 cohort comparison, class-imbalance and ablation experiments, and LASSO- and SHAP-based explainability analyses, are presented and discussed in Chapter 4.

Chapter 4

Implementation and Results

In this chapter, the AI career-anxiety prediction pipeline built as part of the current research is implemented and empirically tested. In particular, it will cover such aspects of the implementation as computational environment, the leak-safe evaluation protocol, and the comparison of the baseline classifiers, class-imbalance treatments, hyperparameter tuning, feature-selection stability methods, explainable-AI (SHAP) importance rankings, feature ablation studies, and ensembling.

4.1 Environment Setup

All the computational workflows were implemented in Python 3.10+ and performed in the Google Colab cloud-based computational environment. The experimental pipeline included eleven consecutive steps, from data loading and cohort construction through benchmarking, class-imbalance handling, hyperparameter optimization, feature-selection stability analysis, explainable-AI (SHAP) importance ranking, feature ablation, ensembling, and final consolidated comparative analysis. At every step, the same global random seed (random_state = 42) was used to ensure that the train/test split, the cross-validation folds, and every stochastic model (e.g., Random Forest, Extra Trees, Gradient Boosting, SMOTE) would be repeatable.

The source data was an anonymized secondary survey dataset, 'Career_Anxiety_due_to_AI.xlsx', downloaded from the Mendeley Data repository. This dataset includes self-reported answers from university students regarding their demographic background, AI tools experience, perceptions of AI impact on employment, and self-reported career anxiety. Out of the fifteen raw columns (participant ID, university, department, age, gender, academic year, AI knowledge, AI tools used, career path, AI-tool perception, AI future perspective, belief that AI will replace jobs, expected AI takeover timeline, career anxiety, and consent), records with missing responses were dropped, and valid participants who were in their 2nd, 3rd, or 4th year of study were retained for the analytical cohorts (from the initial 3,156 valid respondents to 2,036). See Section 4.2 for further details.

Table 4.1: Software and Library Environment Specifications

Component	Role in the Pipeline
Python 3 / Google Colab	Notebook runtime for all computational experiments
pandas, numpy	Data loading, cleaning, regex parsing, and feature engineering

scikit-learn	Preprocessing pipelines, baseline classifiers, cross-validation, evaluation metrics
imbalanced-learn (SMOTE)	Synthetic minority oversampling for class-imbalance sensitivity analysis
SHAP	TreeExplainer-based Shapley value feature-importance attribution
XGBoost	Gradient-boosted decision tree classifier (loaded conditionally)
matplotlib, seaborn	Generation of diagnostic figures and comparative charts
Random state	Fixed at 42 across all splits, estimators, and resampling iterations

4.2 Testing and Evaluation / Performance / Comparative Analysis

The target variable was Anxiety_Label, which was constructed by binarizing the four-level career_anxiety self-reporting variable, mapping 'No Anxiety' and 'Low' to Class 0 (low/no anxiety) and 'Medium' and 'High' to Class 1 (elevated anxiety). For comparative, multi-population testing purposes, four analytical cohorts were constructed from the filtered dataset: Overall (all valid 2nd–4th-year respondents, N = 2,036), Public (subset of public universities, n = 868), Private (private-university subset, n = 1,168), and Daffodil (respondents of the Daffodil International University, n = 695, which is a subset of Private). See Table 4.2 for summary statistics.

Table 4.2: Analytical Cohort Sample Sizes and Class Distribution

Cohort	N	Class 0 (Low/No)	Class 1 (Elevated)	Class 1 (%)
Overall	2,036	660	1,376	67.58%
Public	868	284	584	67.28%
Private	1,168	376	792	67.81%
Daffodil	695	211	484	69.64%

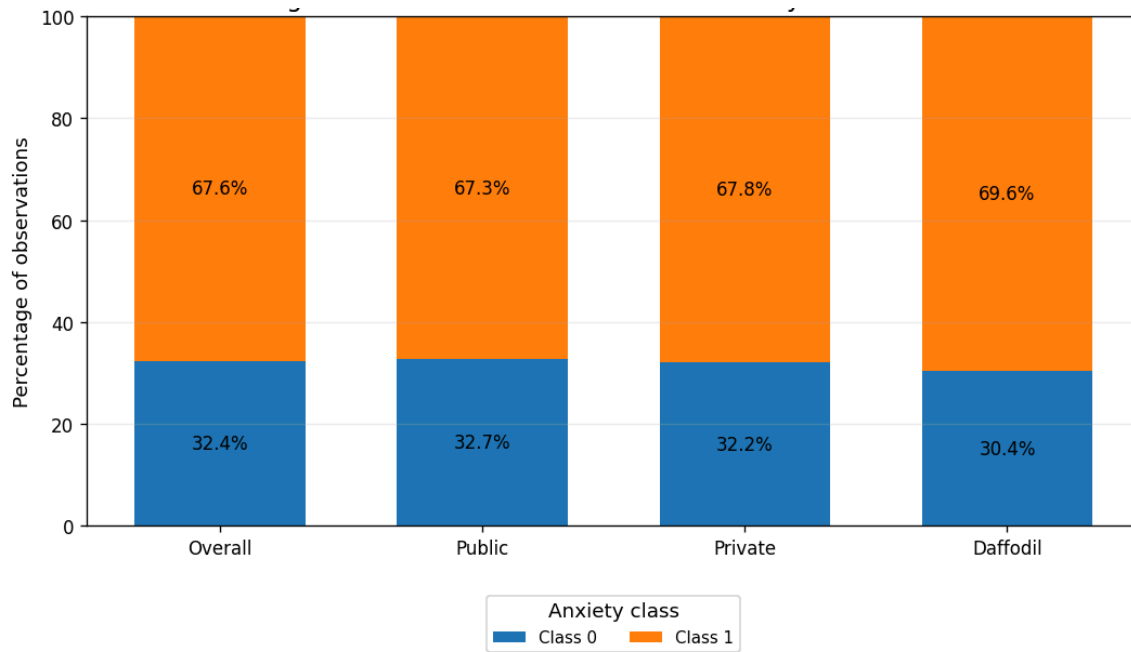


Fig. 4.1. Class distribution across analytical cohorts showing consistent ~67/33 majority imbalance.

All four analytical cohorts demonstrate relatively similar level of class imbalance, with around two-thirds of respondents reporting medium or high career anxiety due to AI and around one-third reporting low or no anxiety. Consistent ~67/33 split motivates using of the class-imbalanced evaluation metrics as well as a separate class-imbalance handling experiment (see Section 4.3.2 below).

Seventeen engineered predictor features were used at every cohort and experiment, grouped in three categories: demographic attributes (age, gender, academic_year, department), AI-exposure attributes (ai_knowledge, Total_AI_Tools, Uses_Text_Gen, Uses_Coding_AI, Uses_Creative_AI), and AI-perception/attitude attributes (career_path, ai_replace_jobs, ai_takeover_time, ai_future_perspective, Threat_Perception, Perceived_Urgency, Risk_Knowledge_Gap, Age_Year_Ratio). Two features were categorical and one-hot encoded (department and career_path), while the remaining fifteen features were numerical and scaled.

The leak-safe evaluation protocol was used in all experiments: (1) a stratified 80/20 train/test split (`test_size = 0.20`) was created once per cohort; (2) model selection and tuning were performed using stratified 10-fold cross-validation on the training partition only; (3) all the preprocessing (imputation, scaling, one-hot encoding, and, where applicable, SMOTE) was fit exclusively inside each of the training folds, never on the held-out test set; and (4) the test set was used only for the final performance reporting and never for feature selection or hyperparameter search. Ten base classifiers were used in the primary classification experiments: Logistic Regression, Linear Discriminant Analysis, Decision Tree, Random Forest, Extra Trees, AdaBoost, Gradient Boosting, Support Vector Machine (SVM), K-Nearest Neighbors, and Gaussian Naive Bayes. XGBoost was imported for the computational environment but was not part of the ten-classifier baseline comparison.

Due to the presence of the persistent $\approx 67/33$ class imbalance, seven different evaluation metrics were calculated at every model: Accuracy, Precision, Recall, F1-score, ROC-AUC, Matthews Correlation Coefficient (MCC), and Cohen's Kappa. F1-score was chosen as the primary optimization metric because accuracy alone could be potentially misleading in imbalanced cases, while ROC-AUC, MCC, and Kappa serve as additional checks that are less sensitive to class imbalance.

4.3 Results and Discussion

4.3.1 Baseline Classifier Benchmark and Cross-Cohort Comparison

At the first modeling stage, the benchmarking of all classifiers was performed on the Overall cohort using the leak-safe protocol described in Section 4.2. Support Vector Machine (SVM) showed the strongest result at the baseline, as shown in Table 4.3 below.

Table 4.3: Overall-Cohort Baseline Best Model (SVM) Held-Out Test Performance

Model	Accuracy	Precision	Recall	F1	ROC-AUC	MCC	Kappa
SVM	0.6887	0.6905	0.9783	0.8096	0.6055	0.1442	0.0797

Then, the identical preprocessing, model set, split strategy, and cross-validation configuration were applied independently to every of the four cohorts, and SVM proved to be the best-performing baseline model in all cases (see Table 4.4).

Table 4.4: Baseline SVM Performance Comparison Across Analytical Cohorts

Cohort	Best Model	F1	ROC-AUC	MCC
Overall	SVM	0.810	0.606	0.144
Public	SVM	0.800	0.513	-0.053
Private	SVM	0.821	0.671	0.233
Daffodil	SVM	0.840	0.728	0.363

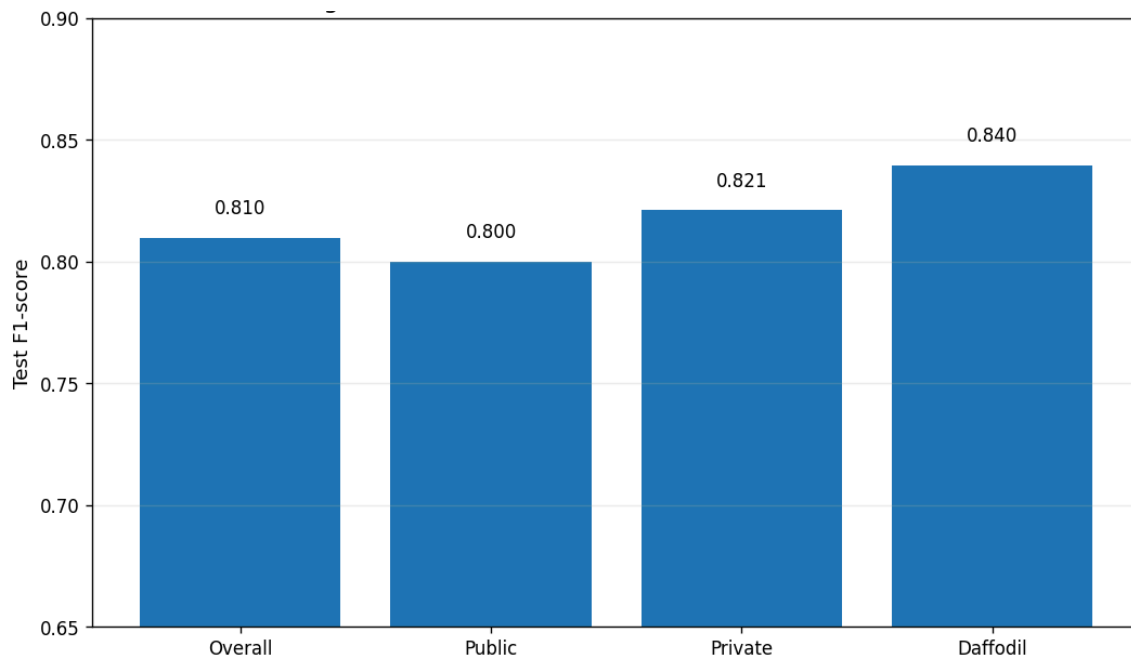


Fig. 4.2. Best baseline F1-score across cohorts.

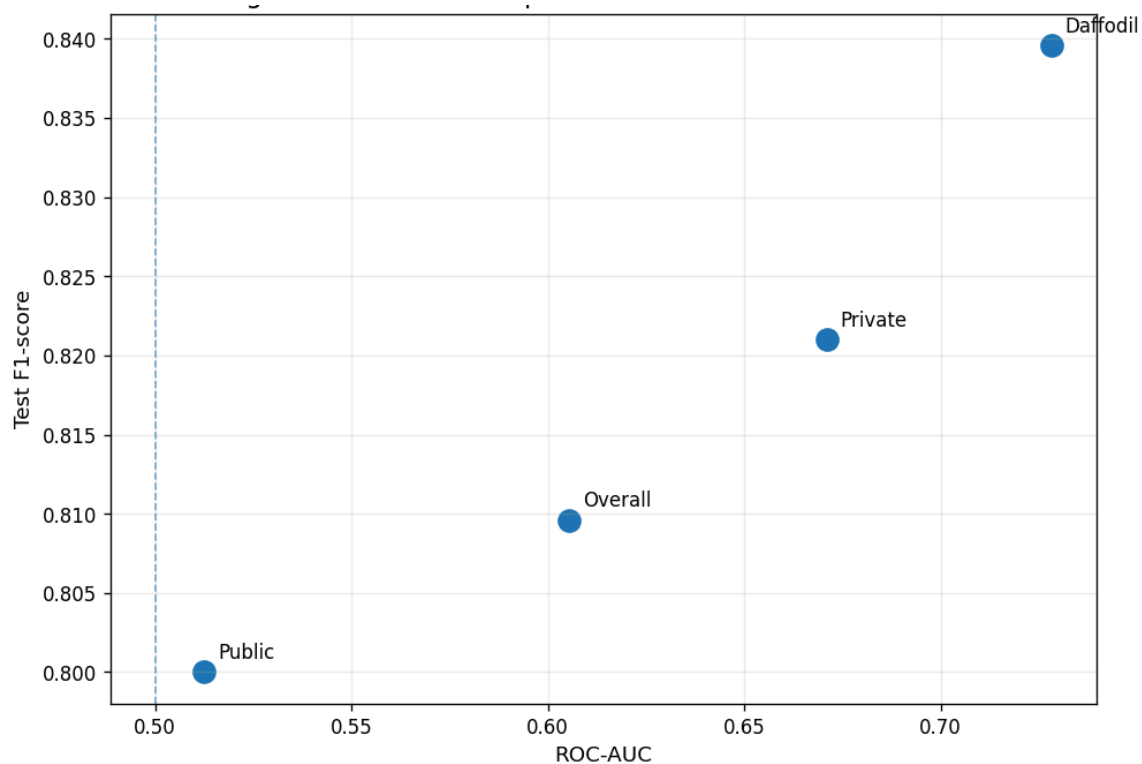


Fig. 4.3. Relationship between F1-score and ROC-AUC for the baseline SVM model.

There are two things to note based on Table 4.4 and Figs. 4.2–4.3. First, F1-scores are consistently high (0.80–0.84) but are mainly driven by very high recall (near 0.97–0.98 for the Overall cohort); this means that the baseline SVM has strong bias towards predicting the majority 'elevated anxiety' class rather than balanced discrimination. Second, ROC-AUC and MCC—which are much less sensitive to class imbalance—vary significantly across cohorts. In

particular, the Daffodil cohort demonstrates the strongest discrimination (ROC-AUC 0.728, MCC 0.363), followed by Private (ROC-AUC 0.671, MCC 0.233), and Public cohort is close to the chance-level discrimination (ROC-AUC 0.513, MCC -0.053). This means that the seventeen engineered features contain more predictive information in the private/Daffodil students population than in the public-university subsample.

4.3.2 Effect of Class-Imbalance Handling Strategies

As F1 at the baseline experiment is biased by the recall of the majority class, a dedicated class-imbalance handling experiment was conducted to compare three conditions—Baseline, Class Weighting, and SMOTE—with the same split, cross-validation, and classifier set, and with SMOTE oversampling performed strictly inside the training pipeline (never on the test fold).

Table 4.5: Best Test F1-Score by Cohort Under Each Imbalance-Handling Condition

Cohort	Baseline F1	Class-Weight F1	SMOTE F1
Overall	0.7153	0.7112	0.6502
Public	0.6992	0.6747	0.6138
Private	0.7198	0.7415	0.6607
Daffodil	0.7292	0.7768	0.6780

Table 4.6: Cross-Cohort Condition-Level Averages for Imbalance Strategies

Condition	F1	ROC-AUC	MCC	Kappa
Baseline	0.7159	0.6059	0.1204	0.1106
Class Weight	0.7261	0.6049	0.1474	0.1415
SMOTE	0.6507	0.5944	0.1186	0.1139

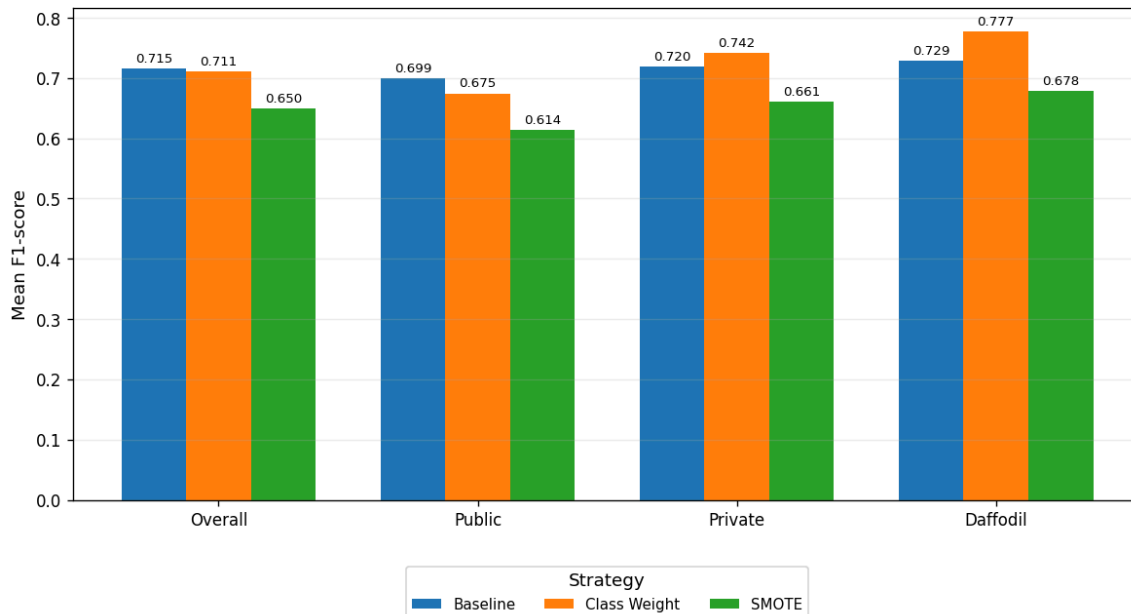


Fig. 4.5. Effect of class-imbalance handling strategies across analytical cohorts.

Class weighting turned out to be the best-performing strategy on the cross-cohort basis (cross-cohort average F1 0.7261 and the best average MCC/Kappa) and especially for the Private (0.7415) and Daffodil (0.7768) cohorts, though slightly decreasing F1 for Overall and Public cohorts in comparison to the unweighted baseline. In turn, SMOTE underperformed both baseline and class weighting on every cohort. Due to SMOTE applied after one-hot encoding of the categorical features at this experiment, its performance should be considered exploratory in the sense that this variant was not compared with SMOTENC, for example. In general, class weighting turned out to be the more reliable imbalance handling strategy and was moved to the next hyperparameter-optimization stage.

4.3.3 Hyperparameter Optimization

The two-stage tuning strategy (a small RandomizedSearchCV pass followed by a focused GridSearchCV pass around the best candidate) was used to tune the two strongest candidates from the previous stage—SVM (baseline) and Random Forest (with class weighting)—using F1 as the main optimization metric under stratified 10-fold cross-validation, with the final performance confirmed on the untouched 80/20 held-out test set.

Table 4.7: Hyperparameter-Optimization Results by Cohort and Model

Cohort	Model	CV F1	Test F1	Test ROC-AUC	Test MCC
Overall	SVM	0.8068	0.8096	0.6055	0.1442
Overall	RF (Class Weight)	0.7141	0.6861	0.5808	0.0445

Public	SVM	0.8045	0.8041	0.5115	0.0000
Public	RF (Class Weight)	0.7093	0.6500	0.4839	-0.1266
Private	SVM	0.8086	0.8072	0.6221	0.0507
Private	RF (Class Weight)	0.7308	0.7171	0.6852	0.1976
Daffodil	SVM	0.8372	0.8396	0.7283	0.3626
Daffodil	RF (Class Weight)	0.7747	0.7821	0.7634	0.4070

Tuned SVM still retained the highest test F1 in every cohort, consistently with the previous observation of the recall-dominated behavior (Section 4.3.1). However, the class-weighted Random Forest achieved higher MCC in the Private (0.1976 vs. 0.0507) and Daffodil (0.4070 vs. 0.3626) cohorts, demonstrating better-balanced discrimination between the two classes despite a lower F1. This F1-vs-MCC trade-off explains why a single metric cannot be sufficient for selecting a classifier under class imbalance; this motivated the multi-model, multi-metric ensemble comparison at Section 4.3.7.

4.3.4 Feature Selection: LASSO Stability Analysis

To find which of the seventeen engineered features contain consistently predictive signal, an L1-regularized logistic regression (LASSO, $C = 0.1$, liblinear solver) was fitted in each of the 10 cross-validation folds per cohort, and a feature was considered 'stable' if it had a non-zero coefficient in a majority of folds.

Table 4.8: LASSO-Stable Feature Sets by Cohort

Cohort	Stable Features (n)	Reduction (%)	Stable Feature Set
Overall	5	70.59	career_path, ai_replace_jobs, ai_takeover_time, ai_future_perspective , Threat_Perception
Public	6	64.71	department, ai_replace_jobs, ai_takeover_time, ai_future_perspective , Uses_Text_Gen,

			Uses_Coding_AI
Private	6	64.71	career_path, gender, Uses_Text_Gen, Threat_Perception, ai_takeover_time, ai_future_perspective
Daffodil	7	58.82	gender, ai_takeover_time, ai_future_perspective, Risk_Knowledge_Gap, Uses_Text_Gen, Perceived_Urgency, Threat_Perception

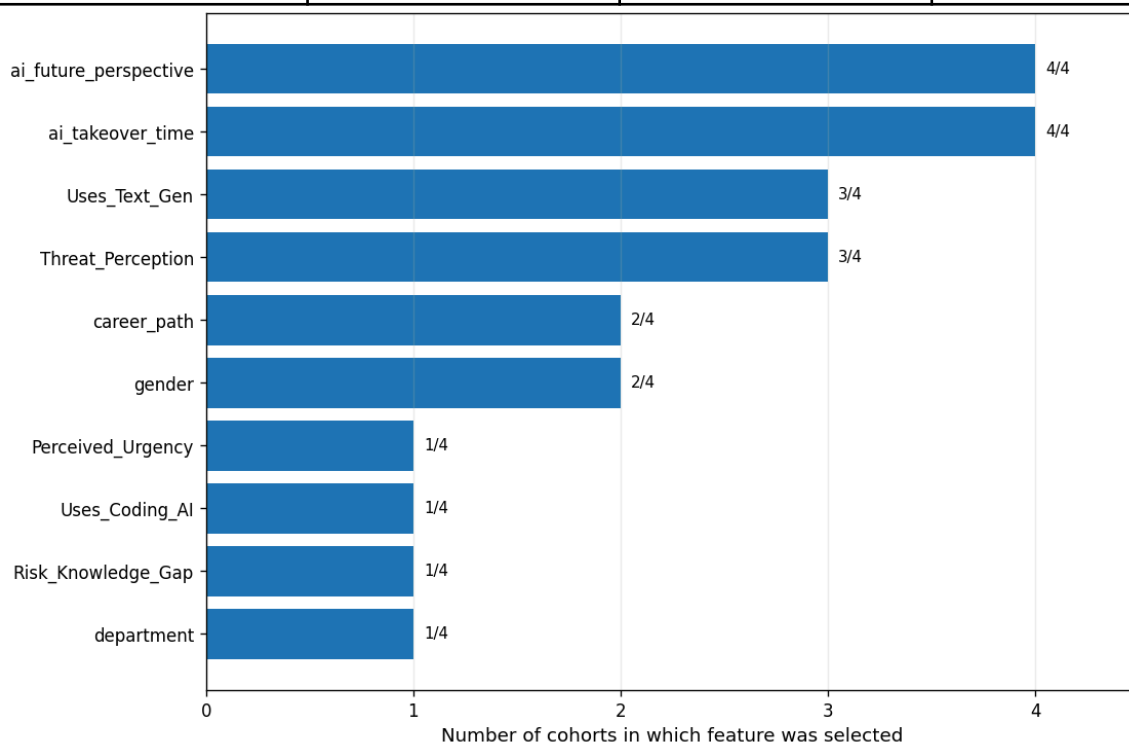


Fig. 4.6. Cross-cohort stability of LASSO-selected features.

Only two features—ai_takeover_time (perceived AI takeover timeline) and ai_future_perspective (belief about AI's future role)—turned out to be selected as stable predictors in all four cohorts, showing that the forward-looking beliefs about AI of the students are the most consistent signal in the dataset. Threat_Perception and Uses_Text_Gen were each stable in three of the four cohorts. LASSO decreased the feature space by 59–71% depending on the cohort.

4.3.5 Feature Importance: SHAP Analysis

A Random Forest (300 estimators, class_weight = 'balanced', random_state = 42) was trained on the training partition of each of the cohorts, and SHAP TreeExplainer values were calculated to rank the top-8 most influential features.

Table 4.9: SHAP Top-8 Features by Cohort

Cohort	SHAP Top-8 Features
Overall	career_path, department, ai_future_perspective, ai_takeover_time, Perceived_Urgency, Risk_Knowledge_Gap, Age_Year_Ratio, age
Public	career_path, department, Age_Year_Ratio, ai_takeover_time, Risk_Knowledge_Gap, age, Perceived_Urgency, ai_future_perspective
Private	career_path, department, ai_takeover_time, Perceived_Urgency, ai_future_perspective, Risk_Knowledge_Gap, Age_Year_Ratio, age
Daffodil	career_path, ai_takeover_time, Perceived_Urgency, department, Risk_Knowledge_Gap, ai_future_perspective, gender, Age_Year_Ratio

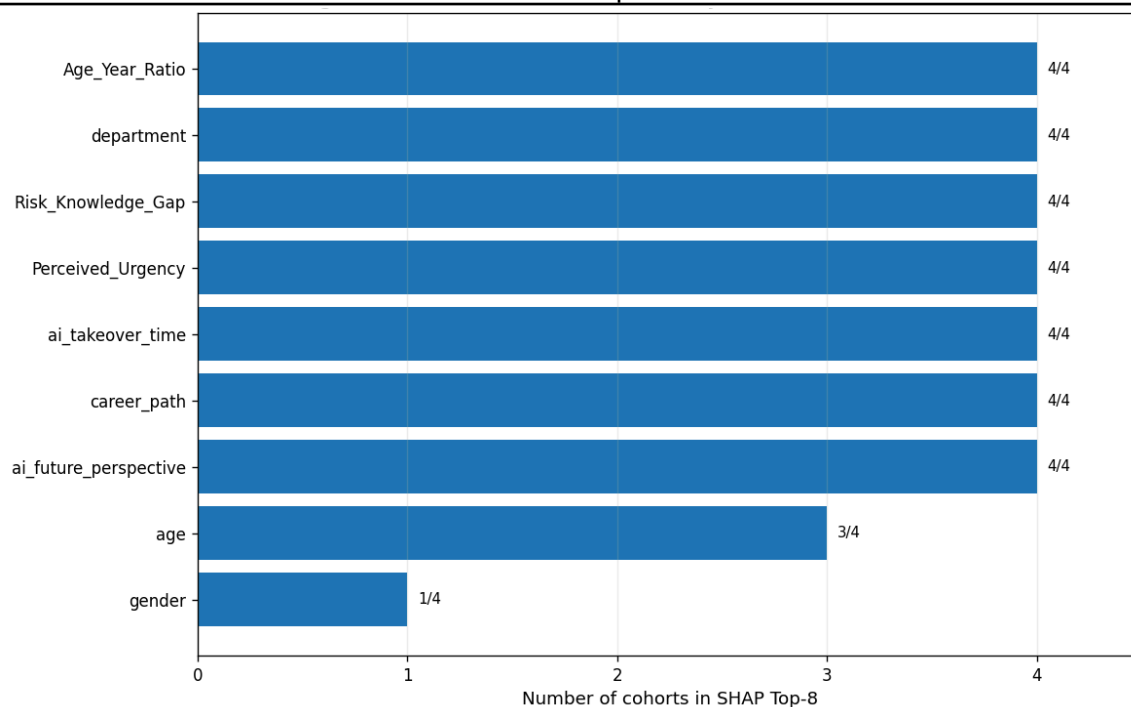


Fig. 4.7. Cross-cohort consistency of SHAP-identified features.

SHAP showed the markedly more consistent set of the features: seven features—career_path, department, ai_future_perspective, ai_takeover_time, Perceived_Urgency, Risk_Knowledge_Gap, and Age_Year_Ratio—entered in the top-8 ranking at all four cohorts, and age entered in three of four. The convergence between a tree-based, non-linear measure of importance (SHAP) and a linear, sparsity-inducing method (LASSO) on ai_takeover_time and ai_future_perspective further validates the interpretation that the outlook of AI and career_path/department context are the dominant factors driving the predicted career anxiety.

4.3.6 Feature Ablation Study

To quantify the cost of feature-space reduction, the three representations of the data—Full 17 features set, the frozen LASSO-stable subset, and the frozen SHAP Top-8 subset—were evaluated with a class-weighted Random Forest (300 estimators) using the same stratified 80/20 split and 10-fold cross-validation.

Table 4.10: Feature Ablation: Full 17 vs. LASSO vs. SHAP Top-8

Cohort	Feature Set	n Features	Reduction (%)	Test F1	Test ROC-AUC	Test MCC	CV F1
Overall	Full17	17	0.00	0.7803	0.5977	0.1048	0.7804
Overall	LASSO	5	70.59	0.6969	0.5480	0.0667	0.6886
Overall	SHAP8	8	52.94	0.7652	0.5842	0.1046	0.7608
Public	Full17	17	0.00	0.7529	0.4846	0.0276	0.7586
Public	LASSO	6	64.71	0.6781	0.4995	0.0260	0.6711
Public	SHAP8	8	52.94	0.7300	0.4793	-0.0724	0.7480
Private	Full17	17	0.00	0.8023	0.6527	0.2334	0.7865
Private	LASSO	6	64.71	0.7233	0.6215	0.1366	0.7043
Private	SHAP8	8	52.94	0.7670	0.6201	0.1672	0.7654
Daffodil	Full17	17	0.00	0.8019	0.7402	0.2405	0.8200
Daffodil	LASSO	7	58.82	0.7487	0.7054	0.1584	0.7481

Daffodil	SHAP8	8	52.94	0.7800	0.7269	0.2190	0.8092
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Figure 4.4. Cohort-Level Predictability Using the Full 17-Feature Representation

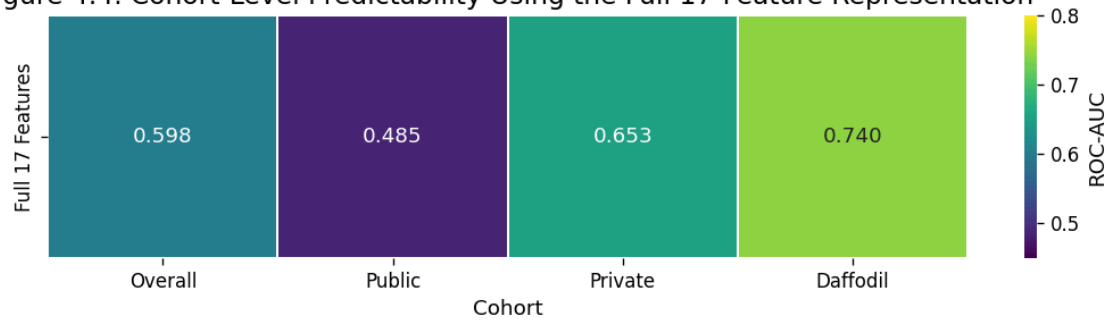


Fig. 4.4. Cohort-level predictability using the Full 17-feature representation.

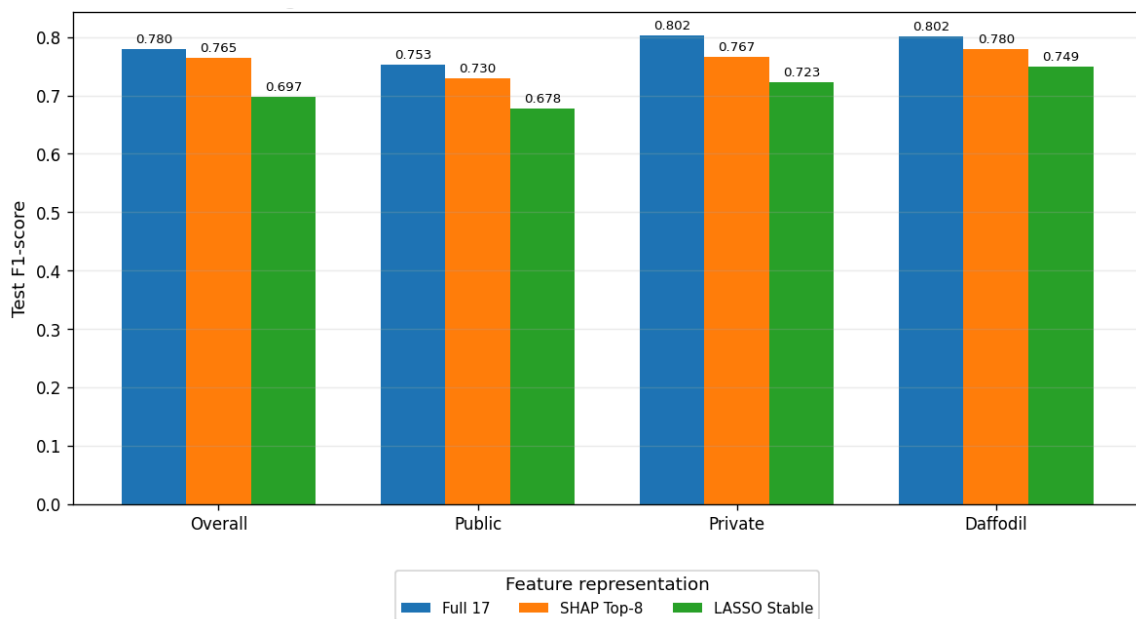


Fig. 4.8. Predictive performance under feature reduction.

Table 4.11: Feature-Set Cross-Cohort Averages

Feature Set	Mean Test F1	Mean Test ROC-AUC	Mean Test MCC	Mean CV F1	Mean Reduction (%)
Full17	0.7844	0.6188	0.1516	0.7864	0.00
SHAP8	0.7606	0.6026	0.1046	0.7708	52.94
LASSO	0.7118	0.5936	0.0969	0.7030	64.71

The Full 17-feature set turned out to have the best average performance, as expected. However, the SHAP Top-8 set retained 96.9% of the Full17 mean F1 (0.7606 vs. 0.7844) and 97.4% of the mean ROC-AUC while dropping the feature space by roughly 53%, a favorable

performance-to-compactness trade-off, and even closer to Full17 specifically at the Daffodil cohort. The LASSO-stable set, despite a slightly larger reduction of the feature space (64.7%), incurred much larger performance loss (mean F1 0.7118, or roughly 90.7% of Full17) in particular at Overall and Public cohorts. This suggests that SHAP-guided feature compacting is the better choice when a smaller, more interpretable set of features is required for downstream deployment/reporting.

Table 4.10 also reproduces the cohort predictability pattern seen at the baseline results: using the Full 17 features set, Daffodil (ROC-AUC 0.7402) and Private (ROC-AUC 0.6527) turned out to be the most predictable cohorts, while Public (ROC-AUC 0.4846, essentially indistinguishable from the chance) remains the least predictable, as shown in Fig. 4.4.

4.3.7 Ensemble Learning

The last modeling stage compared four strong individual classifiers (Random Forest, Extra Trees, Gradient Boosting, Logistic Regression) against four voting ensembles—Performance Hard Voting, Performance Soft Voting, Diverse Hard Voting, and Diverse Soft Voting—built from these base learners using the same leak-safe protocol (no SMOTE in this stage).

Table 4.12: Individual and Ensemble Model Performance by Cohort

Cohort	Model	Type	Test F1	Test ROC-AUC	Test MCC	CV F1
Overall	Random Forest	Individual	0.7771	0.5923	0.0896	0.7831
Overall	Extra Trees	Individual	0.7664	0.5876	0.1132	0.7703
Overall	Gradient Boosting	Individual	0.7994	0.5759	0.0984	0.7992
Overall	Logistic Regression	Individual	0.6641	0.5601	0.0777	0.6701
Overall	Performance Hard Voting	Ensemble	0.7769	0.5407	0.1067	0.7755
Overall	Performance Soft Voting	Ensemble	0.7899	0.6005	0.1347	0.7862
Overall	Diverse Hard	Ensemble	0.7591	0.5379	0.0901	0.7607

	Voting					
Overall	Diverse Soft Voting	Ensemble	0.7841	0.5962	0.1164	0.7798
Public	Random Forest	Individual	0.7640	0.5049	0.0404	0.7465
Public	Extra Trees	Individual	0.7344	0.5276	0.0163	0.7355
Public	Gradient Boosting	Individual	0.7758	0.5199	-0.0671	0.7839
Public	Logistic Regression	Individual	0.5849	0.5019	-0.0462	0.6254
Public	Performance Hard Voting	Ensemble	0.7336	0.4937	-0.0153	0.7419
Public	Performance Soft Voting	Ensemble	0.7537	0.5080	-0.0193	0.7549
Public	Diverse Hard Voting	Ensemble	0.7059	0.4636	-0.0844	0.7339
Public	Diverse Soft Voting	Ensemble	0.7444	0.5010	-0.0415	0.7551
Private	Random Forest	Individual	0.8034	0.6522	0.2294	0.7689
Private	Extra Trees	Individual	0.7714	0.6371	0.1234	0.7614
Private	Gradient Boosting	Individual	0.8065	0.6841	0.1943	0.7945
Private	Logistic Regression	Individual	0.7148	0.6898	0.2642	0.6812
Private	Performance Hard	Ensemble	0.7794	0.5677	0.1616	0.7597

	Voting					
Private	Performance Soft Voting	Ensemble	0.7867	0.6620	0.1264	0.7760
Private	Diverse Hard Voting	Ensemble	0.7868	0.6253	0.2678	0.7508
Private	Diverse Soft Voting	Ensemble	0.7943	0.6812	0.2180	0.7631
Daffodil	Random Forest	Individual	0.8000	0.7561	0.2496	0.8250
Daffodil	Extra Trees	Individual	0.7921	0.7266	0.2452	0.8135
Daffodil	Gradient Boosting	Individual	0.7788	0.7354	0.1383	0.8182
Daffodil	Logistic Regression	Individual	0.7543	0.7553	0.3652	0.7464
Daffodil	Performance Hard Voting	Ensemble	0.7817	0.6231	0.2516	0.8263
Daffodil	Performance Soft Voting	Ensemble	0.8019	0.7418	0.2405	0.8298
Daffodil	Diverse Hard Voting	Ensemble	0.7857	0.6350	0.2739	0.8199
Daffodil	Diverse Soft Voting	Ensemble	0.7822	0.7563	0.2087	0.8321

Table 4.13: Best Individual Model vs. Best Ensemble, by Cohort

Cohort	Best Individual	F1 (Ind.)	Best Ensemble	F1 (Ens.)	Δ F1	Δ ROC-AUC	Δ MCC
--------	-----------------	-----------	---------------	-----------	-------------	------------------	--------------

Overall	Gradient Boosting	0.7994	Performance Soft Voting	0.7899	-0.0095	+0.0246	+0.0363
Public	Gradient Boosting	0.7758	Performance Soft Voting	0.7537	-0.0221	-0.0119	+0.0478
Private	Gradient Boosting	0.8065	Diverse Soft Voting	0.7943	-0.0122	-0.0029	+0.0237
Daffodil	Random Forest	0.8000	Performance Soft Voting	0.8019	+0.0019	-0.0143	-0.0091

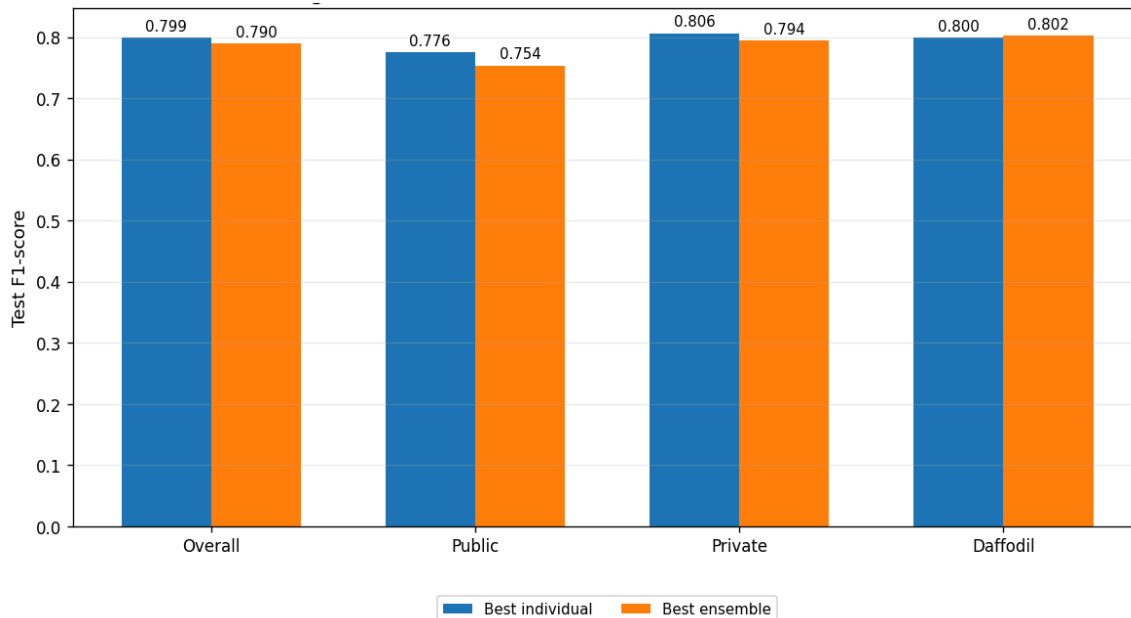


Fig. 4.9. Best individual model versus best ensemble, by cohort.

Ensembling did not consistently outperform the strongest individual model on F1: the best individual model (Gradient Boosting in three of four cohorts, Random Forest in Daffodil) achieved the higher or equal test F1 than the best ensemble in three of the four cohorts, with Daffodil being the sole exception (+0.0019 F1 for Performance Soft Voting). Where ensembling helped, it helped on ROC-AUC and MCC, not on F1—for example, the Public cohort got a +0.0478 MCC gain from ensembling despite a small F1 loss, showing a more balanced decision boundary. Among the four ensemble configurations, Soft Voting (both Performance- and Diversity-based) performed consistently better than Hard Voting in F1 and gave probability-based ROC-AUC scores because hard-voting ensembles produce only discrete class predictions.

4.3.8 Final Comparative Analysis

Step 11 consolidated the frozen results of Steps 3–10 (no model retraining) into cross-cohort summaries to identify the single best-performing model, the most and least predictable cohorts, and the most efficient feature representation.

Table 4.14: Cross-Cohort Model Ranking (Mean Across the Four Cohorts)

Rank	Model	Type	Mean Test F1	Mean ROC-AUC	Mean MCC	Mean CV F1
1	Gradient Boosting	Individual	0.7901	0.6288	0.0910	0.7990
2	Random Forest	Individual	0.7861	0.6264	0.1522	0.7809
3	Performance Soft Voting	Ensemble	0.7830	0.6281	0.1206	0.7867
4	Diverse Soft Voting	Ensemble	0.7762	0.6337	0.1254	0.7825
5	Performance Hard Voting	Ensemble	0.7679	0.5563	0.1261	0.7758
6	Extra Trees	Individual	0.7661	0.6197	0.1245	0.7702
7	Diverse Hard Voting	Ensemble	0.7594	0.5654	0.1368	0.7663
8	Logistic Regression	Individual	0.6795	0.6268	0.1652	0.6808

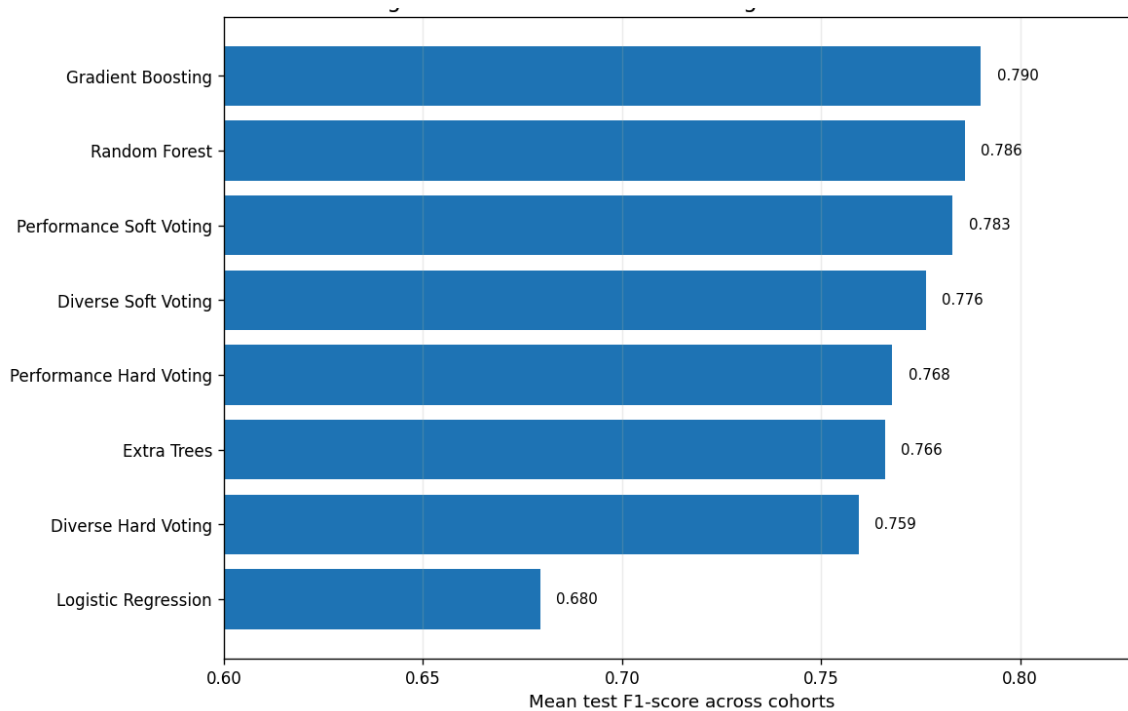


Fig. 4.10. Cross-cohort ranking of predictive models.

Table 4.15: Cohort Predictability Under the Full 17-Features Representation

Rank	Cohort	Test F1	Test ROC-AUC	Test MCC	Test Kappa	CV F1	CV ROC-AUC
1	Daffodil	0.8019	0.7402	0.2405	0.2333	0.8200	0.7302
2	Private	0.8023	0.6527	0.2334	0.2135	0.7865	0.6325
3	Overall	0.7803	0.5977	0.1048	0.0907	0.7804	0.5857
4	Public	0.7529	0.4846	0.0276	0.0248	0.7586	0.5141

Gradient Boosting turned out to have the best cross-cohort mean test F1 (0.7901), CV F1 (0.7990) and narrowly beat Random Forest (mean F1 0.7861), though Random Forest achieved a noticeably higher mean MCC (0.1522 vs. 0.0910), again reflecting the F1 vs. MCC trade-off seen in Section 4.3.3. By cohort predictability, Daffodil is confirmed as the most discriminating cohort (ROC-AUC 0.7402, MCC 0.2405), and Public the least (ROC-AUC 0.4846, essentially the chance-level), as seen at every experiment in this chapter.

Table 4.16: Feature Consistency Across Cohorts (LASSO vs. SHAP)

Feature	LASSO Cohort Count (/4)	SHAP Cohort Count (/4)

ai_takeover_time	4	4
ai_future_perspective	4	4
career_path	2	4
department	1	4
Perceived_Urgency	1	4
Risk_Knowledge_Gap	1	4
Age_Year_Ratio	0	4
age	0	3
Threat_Perception	3	0
Uses_Text_Gen	3	0
gender	2	1
ai_replace_jobs	2	0
Uses_Coding_AI	1	0

Table 4.17: Methodological Interpretation Notes

Issue	Status	Interpretation
Class imbalance	Important	Class-weighting improved cross-cohort condition-level performance relative to the unweighted baseline.
SMOTE	Exploratory only	Step 5 SMOTE was applied after one-hot encoding, so it should not be treated as a definitive categorical-safe SMOTE result.
LASSO	Feature reduction	Stable LASSO subsets reduced dimensionality substantially but caused a noticeable

		predictive-performance loss.
SHAP	Compact representation	SHAP Top-8 retained substantially more performance than LASSO and remained close to Full17 in the Daffodil cohort.
Ensemble	Metric-dependent	Ensembles did not consistently outperform the strongest individual model in F1.
Soft voting	Preferred ensemble	Soft voting generally outperformed hard voting in F1 and produced more meaningful, probability-based ROC-AUC.
Public cohort	Weak generalization	Public-cohort ROC-AUC remained close to chance level across several analyses.
Daffodil cohort	Strongest discrimination	Daffodil repeatedly produced the strongest ROC-AUC/MCC profile among the four cohorts.

Combining the eleven-step pipeline, four consolidated conclusions could be drawn. First, the best cross-cohort model is Gradient Boosting, with mean test F1 0.7901, mean ROC-AUC 0.6288 and mean CV F1 0.7990, though Random Forest can be a better-balanced alternative on MCC. Second, cohort predictability is not uniform: Daffodil is the most discriminating population (ROC-AUC 0.7402) and Public the least (ROC-AUC 0.4846), gap of around 26 ROC-AUC points which is stable across baseline, tuning, ablation, and ensembling experiments. Third, the SHAP Top-8 feature set is the most defensible compact representation, retaining ~97% of Full17's mean F1 while dropping 53% of the feature space, and it is also far more consistent across cohorts than LASSO-stable subset. Finally, only two features—`ai_takeover_time` and `ai_future_perspective`—were found important by both a linear (LASSO) and non-linear (SHAP) methods in all four cohorts, thus showing that students' beliefs about the speed and inevitability of AI-driven job displacement are the single most robust predictor of career anxiety in this dataset.

4.4 Summary

This chapter implemented and evaluated an eleven-step machine learning pipeline for

predicting AI-induced career anxiety among university students across four analytical cohorts (Overall, Public, Private, and Daffodil). Using the leak-safe stratified 80/20 split with 10-fold cross-validation and seven evaluation metrics, the baseline SVM classifier achieved F1-scores of 0.80–0.84 but moderate discrimination (ROC-AUC 0.51–0.73), a gap which is explained by the persistent $\approx 67/33$ class imbalance in the target variable. Class weighting turned out to be a more reliable imbalance-handling strategy than SMOTE. The LASSO and SHAP feature selection techniques converged on `ai_takeover_time` and `ai_future_perspective` as the two most consistently important predictors across all cohorts, and the SHAP Top-8 feature set offered the best trade-off between feature-space compactness and performance retention. Voting ensembles did not consistently outperform the strongest individual model (Gradient Boosting) on F1, though soft-voting ensembles improved MCC in several cohorts. At every experiment, the Daffodil cohort was the most predictable, and the Public the least, showing that model performance—and, therefore, the strength of feature-anxiety relationship—is heterogeneous across the student populations. These results, together with the engineering standards in Chapter 5, informed the conclusions and limitations in Chapter 6.

Chapter 5

Engineering Standards and Design Challenges

In this chapter, the engineering standards, design constraints, impact on society, economic viability, and Complex Engineering Problem (CEP) of the career-anxiety predictive modeling system for artificial intelligence practitioners are discussed. Furthermore, the computational pipeline will be mapped to the Washington Accord and Board of Accreditation for Engineering and Technical Education (BAETE) problem-solving and activities dimensions.

5.1 Compliance with the Standards

The engineering standards define the baseline parameters for reproducibility and isolation of code implementation, cryptographic isolation and reliable computing. In designing the machine learning architecture for student career anxiety prediction, the international and industry standards have been compared to create an auditable and secure software lifecycle.

5.1.1 Software Standards

The software architecture has been developed according to ISO/IEC 25010 (Software Quality Models), IEEE 730-2014 (Standard for Software Quality Assurance Processes), IEEE 829-2008 / IEEE 29119 (Software Testing Documentation) and Python PEP 8 code style guide.

1. ISO/IEC 25010 (Software Quality Requirements and Evaluation): Governing the functional suitability and architectural modularity. The analytical pipeline strictly isolates numerical scaling and categorical one-hot encoding into scikit-learn Pipeline and ColumnTransformer abstraction layers. The parameters for imputation (median for numerical attributes and mode for categorical attributes) and scaling (parameters) are calculated only for training folds and therefore, there is absolutely no data leakage into validation and testing partitions among the 2,036 surveyed student records. Beyond functional suitability, the standard's Usability characteristic is addressed by the demonstration Streamlit interface described in Chapter 3, which exposes the frozen pipeline through a structured, form-based input flow and an explicit prediction/explanation view, so that a user can obtain and interpret a prediction without directly interacting with the underlying notebook or code.

Analysis of Alternatives:

- **Alternatives Considered:** Global transformations of data with scripts and independent normalization functions.
- **Pros and Cons:** While independent scripts provide an ability to experiment with the model in an exploratory manner, they cause contamination of the test partition with global sample distributions and thus, produce overestimated and non-generalizable metrics. The Pipeline standardization introduces a small abstract layer but assures

mathematical isolation of training and test datasets.

- **Rationale of Selection:** The ISO/IEC 25010 standard pipeline is crucial for institutional machine learning systems in order to obtain the accurate performance metrics of the model which generalizes on new student cohorts.

2. IEEE 829-2008 / IEEE 29119 (Software Testing Standards): Governing the verification of experiments and rigorous validation. The empirical framework ensures the usage of the Stratified 10-Fold Cross-Validation scheme ($k=10$, $\text{shuffle}=\text{True}$, $\text{random_state}=42$) and untouched 80/20 hold-out testing partition. The stratification rigorously maintains the proportion of the target class (67.58% anxious vs. 32.42% non-anxious).

Analysis of Alternatives:

- **Alternatives Considered:** Non-stratified k-fold cross-validation, simple train/test holdout partitioning and leave-one-out cross-validation (LOOCV).
- **Pros and Cons:** In case of non-stratified partitioning, the minority class proportion may vary in minority partitions and thus, the evaluation becomes unstable. LOOCV produces prohibitively computationally expensive procedure on 2,036 observations without decreasing evaluation variance. The stratified 10-fold CV provides optimal bias-variance balance and statistical stability.
- **Rationale of Selection:** The IEEE testing standards require the reproducible and statistically sound validation. The stratified 10-fold CV has proven that the best performing classifiers (SVM with CV F1 of 0.8068 ± 0.0073 and AdaBoost with CV F1 of 0.8028 ± 0.0052) are robust and not prone to fold-dependent overfitting.

3. PEP 8 & Open Source Machine Learning Standards: Governing the readability and consistency of the Python codebase as well as deterministic execution. Deterministic pseudo-random seed initialization ($\text{seed}=42$) has been applied across the NumPy, scikit-learn and tree ensemble models to assure the multi-platform bit-level reproducibility.

5.1.2 Hardware Standards

The development, benchmark execution locally and cloud scaling conformed to IEEE 1621 (Standard for User Interface Elements in Power Control) and IEC 62368-1 (Safety requirements for audio/video, information and communication technology equipment).

Energy-Efficient Edge Workstation Architecture: The local development and execution of the model has been performed on the Apple Silicon M2 System-on-Chip (ARM64 architecture, 8-core CPU and unified LPDDR5 memory subsystem with 100 GB/s bandwidth).

Analysis of Alternatives:

- **Alternatives Considered:** High-TDP discrete GPU workstations (e.g., x86_64 platforms with dedicated NVIDIA RTX 3080/4090 GPUs).
- **Pros and Cons:** The discrete GPU servers provide massive parallelism in the form of floating point computations for deep tensor networks but consume 350 W to 600 W of

energy and produce substantial thermal loads which require the continuous mechanical cooling. The ARM64 unified memory architecture provides the high cache coherence and memory throughput with the average power budget under 30 W.

- **Rationale of Selection:** For the tabular machine learning with 17 engineered features, the computation bottleneck is memory bandwidth rather than tensor-computations. The low-TDP ARM system reduced both the carbon footprint and cost of hardware while assuring the fast model training time of sub-minute.

Cloud Virtualization Standards: The model validation and hyper-parameter tuning in cloud have been performed using the ephemeral instances deployed in the ISO/IEC 27001 certified data centers (Google Colab infrastructure) which provide the isolated compute sandboxes with secured runtime environments.

5.1.3 Communication Standards

The present study uses a secondary survey dataset. A demonstration Streamlit interface for the trained models was subsequently developed (Chapter 3), but it currently runs locally and is not publicly deployed or exposed to network traffic; communication standards therefore remain primarily relevant to any future public web-based deployment and digital data transfer. IETF RFC 9110 / RFC 8446 (HTTP Semantics and TLS 1.3): The present study did not conduct primary online data collection. If the demonstration interface described in Chapter 3 is deployed publicly in the future, HTTPS with TLS should be used to provide confidentiality and integrity protection in transit.

Analysis of Alternatives:

- **Alternatives Considered:** Unencrypted HTTP transfer or legacy TLS 1.0/1.1.
- **Pros and Cons:** The legacy and unencrypted protocols would make the sensitive student demographic records, academic identifiers and psychological perceptions vulnerable to packet inspection and man-in-the-middle (MITM) attacks. TLS 1.3 eliminates the vulnerable cryptography and reduces the connection handshake latency.
- **Rationale of Selection:** The enforcement of TLS 1.3 protocol satisfies the standard institutional research ethics requirement for participant confidentiality in transit.

IETF RFC 8259 (JSON) and ISO/IEC 29500 (OpenXML): All dataset states and model artifact metadata have been standardized into OpenXML (.xlsx) and JSON formats, thus removing any byte-order dependency and allowing for the seamless ingestion in diverse computing environments.

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

The deployment of generative AI technologies in recent years has already caused serious disruptions in higher education student careers [1, 2]. Among the 2,036 analytical records from the participating public and private university cohorts, 67.58% (1,376 students) were classified as having medium or high career anxiety associated with AI-related career change, whereas only 32.42% (660 students) have low levels of anxiety.

The current study provides an automated empirical testbed for predicting career anxiety among higher education students. With the help of the developed predictive system, the Risk-Knowledge Gap and Perceived Urgency of obsolescence become the main indicators, which can guide the university counseling centers and career advisors in implementing specific interventions and career resiliency workshops before the student suffers from the career despair and dropouts [1, 4].

5.2.2 Impact on Society & Environment

Societal Implications: The shift towards the knowledge economy powered by AI is expected to increase the social inequality between students possessing the literacy in AI technology and those threatened by structural unemployment [3]. The developed system allows educational administrators, curricula developers and national policy-makers to obtain detailed knowledge on student vulnerability across different subjects. The degree requirements in universities like Daffodil International University can be adjusted accordingly and prevent the societal technostress [15, 23].

Environmental Implications: The machine learning research today is usually performed using the deep learning models, requiring huge amounts of electrical power. This work conforms to the Green Computing approach and considers the systematic analysis of very efficient classical machine learning algorithms (Logistic Regression, Linear Discriminant Analysis, Decision Trees, Random Forests, Extra Trees, AdaBoost, Gradient Boosting, Support Vector Machines, K-Nearest Neighbors, Gaussian Naive Bayes). The complete training and cross-validation procedure is performed within seconds and has relatively modest computational requirements compared with large-scale deep-learning workloads.

5.2.3 Ethical Aspects

5.2.4

The integrity of the project is assured during all stages of the research:

4. **Informed Consent:** The secondary dataset contains a consent field, and all 3,156 records in the provided dataset were marked as consent = 'Yes'. No primary participant recruitment or survey administration was conducted as part of this study.
5. **Data Anonymization:** The institutional unique student identifiers and tracking strings (participant_id) have been totally isolated from the feature set to exclude the re-identification risk.

6. **Algorithmic Fairness:** Predictive behavior was compared across the analyzed institutional cohorts and available demographic groups to identify performance differences. A formal algorithmic fairness audit was outside the scope of this study.
7. **Interpretability:** The decision mechanisms of the model have been audited using the Explainable AI (SHAP) principles rather than being treated as a black box and thus, are verifiable by educators and students.

5.2.5 Sustainability Plan

Three pillars of sustainability have been implemented into the system:

- **Open-Source Maintainability:** The codebase utilizes only the permissive open-source libraries (scikit-learn, pandas, numpy, imbalanced-learn, shap), whose versions are strictly pinned to ensure the forward compatibility without additional costs.
- **Longitudinal Extensibility:** The modular architecture of the ColumnTransformer allows institutional researchers to add new longitudinal cohorts of students to the engine as soon as generative AI technologies will evolve further, calibrate the decision boundary accordingly but not redesign the software pipeline.
- **Institutional Viability:** Without any software licensing fees and low-computational requirements, universities in developing countries could maintain and run the prediction pipeline with relatively low software and computational overhead on the existing institutional IT infrastructure.

5.3 Project Management and Financial Analysis

An estimated budget analysis was conducted to assess the financial feasibility of a potential academic implementation and future deployment of the career-anxiety prediction system in educational institutions. The comparison of the proposed lean academic implementation of the system with the enterprise-grade cloud deployment is provided in Table 5.0.

Table : 5.0. Comprehensive Financial and Budgetary Analysis

Expenditure Category	Proposed Academic Budget (BDT)	Lean Budget	Alternate Enterprise Budget (BDT)	Rationale & Trade-off Analysis

Compute & Development Hardware	BDT 165,000	BDT 450,000	The lean configuration assumes an academic workstation or equivalent compute access sufficient for classical tabular machine-learning experiments; the enterprise configuration allows for higher-capacity infrastructure and associated maintenance.
Cloud Computing & Sandbox	BDT 0 (Free/Academic Tier)	BDT 120,000 (AWS/GCP)	The lean configuration relies on available free or academic cloud-computing resources; the enterprise configuration assumes persistent managed cloud infrastructure and machine-learning services.
Data Acquisition	BDT 0	BDT 0	The present study uses a secondary survey dataset obtained from the Mendeley Data repository. No participant recruitment, survey administration, or primary data-collection expenditure was incurred by the research team.
Software Licensing & Tooling	BDT 0 (Open-Source Stack)	BDT 95,000 (Commercial Software)	The research pipeline uses Python and open-source machine-learning libraries, avoiding commercial licensing costs. An enterprise configuration may include commercial statistical, visualization, monitoring, or analytics software.

Domain, Hosting & API Serving	BDT 6,000	BDT 45,000	A lightweight future demonstration can use low-cost or institutional hosting; enterprise deployment may require managed containers, monitoring, security controls, and service-level guarantees.
Contingency & Audit (10%)	BDT 17,100	BDT 71,000	Provision for unexpected hardware, cloud, deployment, security, documentation, and audit-related expenses; calculated at approximately 10% of the corresponding non-contingency budget.
Total Estimated Budget	BDT 188,100	BDT 781,000	These are estimated implementation/deployment budgets rather than actual research expenditure. The lean configuration minimizes infrastructure and licensing requirements while remaining appropriate for the current research workload.

Although initiated within an undergraduate design-project framework, the proposed system could support the following future institutional services and sustainability model:

Institutional SaaS Subscription Licensing: Educational institutions could potentially subscribe to a validated predictive analytics dashboard on an annual basis to monitor the longitudinal trends of student anxiety, analyze the effectiveness of curriculum changes, and benchmark student well-being indices against the national average.

Consulting Services: Aggregated, non-identifying findings could potentially provide consulting value to university administrators, vocational training boards, and corporate recruiting companies trying to understand the workforce skill mismatch in the age of AI.

Grant Writing & Research Commercialization: The empirically established prevalence rate

(67.58% of anxiety among 2,036 undergraduate respondents) could provide a basis for future grant applications to conduct further research nationally and internationally on technological employment resilience and mental healthcare in universities.

5.4 Complex Engineering Problem

The formulation, modeling, and implementation of the AI-related career-anxiety prediction pipeline is a Complex Engineering Problem (CEP) according to the criteria of the Washington Accord and BAETE guidelines.

5.4.1 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.1). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.1: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledg e	EP2 Range Of Conflicting Requirement s	EP3 Depth of Analysi s	EP4 Familiarit y of Issues	EP5 Extent of Applicabl e Codes	EP6 Extent Of Stake- holder Involveme nt	EP7 Interdependen ce
✓	✓	✓	✓			✓

EP1 (Depth of Knowledge): The project requires integrated knowledge of machine learning, statistical learning, feature engineering, regular-expression-based text parsing, categorical encoding, model evaluation, and explainable AI techniques to develop and evaluate the predictive pipeline. [3.1.2, 4.3.4, 4.3.5]

EP2 (Range of Conflicting Requirements): The study involves balancing competing evaluation requirements, particularly achieving high recall for the elevated-anxiety class while maintaining balanced discrimination and limiting false-positive predictions, as reflected by the use of F1-score, MCC, and Cohen's Kappa. [3.1.2, 4.3.1, 4.3.3]

EP3 (Depth of Analysis): The project requires comparative analysis of multiple classification algorithms, class-imbalance strategies, feature-selection methods, SHAP-based interpretation, feature ablation, and ensemble learning across four analytical cohorts. [3.2, 4.3.1–4.3.8]

EP4 (Familiarity of Issues): The problem involves the emerging and rapidly changing effects of AI on students' perceived career prospects, requiring adaptation of established machine-learning techniques to a relatively new research context. [2.2, 2.3]

EP7 (Interdependence): The stages of data preparation, target construction, feature engineering, leak-safe preprocessing, model training, feature selection, explainability, and final evaluation are interdependent; errors in earlier stages can affect all subsequent results. [3.1.1, 3.3, 4.1]

Mapping with Knowledge Profile

This section is designed to map the overall problem and EP1 (*multiple between K3, K4, K5, K6, K8 for attaining EP1*) to the Knowledge Profile.

Table 5.2: Mapping with knowledge Profile.

K1 Natural Science	K2 Mathematics	K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K7 Comprehension	K8 Research Literature
		✓	✓	✓	✓		✓

K3 (Engineering Fundamentals): The project applies fundamental engineering and computing concepts including algorithm design, data processing, computational procedures, and modular pipeline construction. [3.1.1, 3.1.2]

K4 (Specialist Knowledge): The work requires specialist knowledge of supervised machine learning, classification algorithms, ensemble methods, regularization, SHAP-based explainability, and class-imbalance handling. [3.1.2, 3.2, 4.3.5, 4.3.7]

K5 (Engineering Design): A leak-safe 17-feature analytical pipeline was designed to transform the raw survey data into a structured representation suitable for comparative predictive modeling. [3.1.2, 3.1.4, 3.1.5]

K6 (Engineering Practice): The project applies practical engineering methods through Python, scikit-learn, pandas, reproducible random-state control, structured experimentation, and systematic evaluation. [3.1.2, 4.1]

K8 (Research Literature): The study is grounded in relevant research literature on AI-related anxiety, job insecurity, career decision-making, AI self-efficacy, and student experiences with AI, which informs the research design and interpretation of findings. [2.1, 2.2, 2.3]

5.4.2 Engineering Activities

The execution of the research involved the complex and non-routine engineering activities including multi-institutional data integration and non-linear modeling with significant societal consequences.

Mapping with Complex Engineering Activities

The project maps across all five Complex Engineering Activity categories (EA1 through EA5) (see Table 5.3).

Table 5.3: Mapping with Engineering Activities.

EA1 Range of Resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for Society & Environment	EA5 Familiarity

✓	✓			✓
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Rationale for Engineering Activities Mapping:

EA1 (Range of Resources): The project uses a combination of computational resources, open-source Python libraries, Google Colab, the secondary multi-institutional survey dataset, and the analytical notebook required to conduct the full experimental workflow. [4.1, 5.3]

EA2 (Level of Interaction): The project required coordination between the student researchers, academic supervisors, and the institutional requirements of the FYDP process, while the analysis also involved handling data representing multiple university cohorts. [3.3, 3.4]

EA5 (Familiarity): The project addresses an emerging AI-related career-anxiety problem for which no standard off-the-shelf predictive framework was directly available, requiring the adaptation and integration of established machine-learning methods into a new research context. [2.2.1, 2.3]

5.5 Summary

This chapter described the engineering framework of the AI career-anxiety prediction system. Conformity to ISO/IEC 25010, IEEE 730, and applicable secure-communication standards supports the software quality, data isolation and cryptographic security. The societal, ethical and environmental analysis shows that the system targets to solve the real problems of students without causing any environmental pollution or exposing participants' identities. Moreover, the budget analysis shows the lower estimated capital requirement of the lean configuration compared with the enterprise configuration due to more lean computation setup. Finally, the mapping of the project against BAETE and Washington Accord criteria confirms its status as a Complex Engineering Problem (EP1–EP7, K2–K8, EA1–EA5). This will serve as the background for conclusions and future directions presented in Chapter 6.

Chapter 6

Conclusion

This chapter summarizes the key findings of the project, discusses the study's limitations, and outlines potential directions for future research based on the machine-learning framework developed.

6.1 Summary

The rapid adoption of generative artificial intelligence (GenAI) tools across the professional landscape has disrupted the labor market in which the current generation of graduates will have to operate [1], [2]. Technostress and increased difficulty in tailoring career plans has been experienced by many students enrolled in computer science, software development, business administration, food engineering, and healthcare majors [3], [7], [16]. The present Final Year Design Project (FYDP) offers an empirical, leak-safe machine learning approach towards modeling, classification, and interpretation of career anxiety of GenAI era students. The research is based on a wide empirical cross-sectional survey database consisting of 3,156 respondents from undergraduate courses at five of the best Bangladeshi universities, such as Daffodil International University (33.71%, $n = 1,064$), American International University-Bangladesh (21.96%, $n = 693$), University of Dhaka (19.68%, $n = 621$), Jahangirnagar University (17.93%, $n = 566$), and Chittagong University of Engineering and Technology (6.72%, $n = 212$). In compliance with strict analysis requirements regarding the vulnerability of the workforce, the sample set for further analysis was limited to 2nd-, 3rd-, and 4th-year students ($N = 2,036$) by purposeful exclusion of the 1st-year students ($n = 1,120$), who were still at the experimental stage of education, irrespective of career deployment. Targeted formulation has created a robust binary classification system (Anxiety_Label in {0, 1}), with low and negligible levels of anxiety belonging to Class 0 ($n = 660$, 32.42%) and medium and severe career anxiety belonging to Class 1 ($n = 1,376$, 67.58%). This binary classification has demonstrated the existence of a significant baseline prevalence of GenAI-induced anxiety among senior university students in both public and private sectors, where the majority of them (more than two-thirds) suffer from career anxiety because of the AI automation threat. A strict 17-variable analytical framework was designed for capturing technical competency, perceived threat, and domain-specific characteristics without causing data leakage. Alongside demographic control variables (academic year, age, gender, academic department, and future career), ordinal scales of domain-specific indicators were applied for quantifying the level of AI literacy [1], [15], likelihood of substitution for future occupation [8], [12], timeline of GenAI takeover [23], and threat perspective [13]. High-quality lexical regex parsers helped to separate comma-separated lists of software tools for measuring experience with text generators (ChatGPT, Claude, Gemini, QuillBot), code copilots (GitHub Copilot, Cursor, Replit, DeepSeek), and creative synthesis engines (Midjourney, DALL-E, Canva, Photomath). In addition, three composite indicators were developed to measure latent cognitive friction: Perceived Urgency ($ai_replace_jobs * ai_takeover_time$), Risk-Knowledge Gap ($ai_future_perspective - ai_knowledge$), and the Age-Year Ratio ($age / academic_year$). A leak-proof preprocessing pipeline with scikit-learn ColumnTransformer was designed to apply

median imputation, standard scaling, categorical most-frequent imputation, and one-hot encoding of predictors, fitted only on training folds. The analytical sample was split into 80% stratified training data (n = 1,628) and a 20% validation dataset (n = 408). Comparative benchmarking of ten classical, linear, tree-based, and ensemble classifiers based on Stratified 10-Fold Cross-Validation was performed using seven traditional performance evaluation metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC, Matthews Correlation Coefficient (MCC), and Cohen's Kappa. In addition, comparative benchmarking was done with respect to four cohorts: Overall (N = 2,036), Public Universities (N = 868), Private Universities (N = 1,168), and the main institution, Daffodil International University (N = 695).

Table 6.1: Master Baseline Classifier Performance (Stratified 10-Fold CV and Held-Out Test Evaluation)

Model	CV Acc	CV Prec	CV Rec	CV F1	CV AU C	CV MCC	Test Acc	Test F1	Test Rec	Test MCC	Test κ
Support Vector Machine (SVM)	0.6831	0.6861	0.9791	0.8068	0.5640	0.1115	0.6887	0.8096	0.9783	0.1442	0.0797
AdaBoost Classifier	0.6708	0.6743	0.9918	0.8028	0.5918	-0.0263	0.6765	0.8070	1.0000	0.0000	0.0000
Gradient Boosting Classifier	0.6738	0.6873	0.9491	0.7972	0.5986	0.0943	0.6863	0.8061	0.9638	0.1388	0.0888
Logistic Regression	0.6677	0.6900	0.9227	0.7895	0.6004	0.0966	0.6667	0.7914	0.9348	0.0711	0.0509
Random Forest Classifier	0.6548	0.6906	0.8873	0.7764	0.5949	0.0775	0.6691	0.7840	0.8877	0.1328	0.1166
Linear Discriminant Analysis (LDA)	0.6462	0.6859	0.8782	0.7701	0.5936	0.0591	0.6520	0.7788	0.9058	0.0416	0.0329
Extra Trees Classifier	0.6536	0.6988	0.8573	0.7697	0.5939	0.1070	0.6422	0.7622	0.8478	0.0744	0.0683
K-Nearest Neighbors (KNN)	0.6303	0.6932	0.8127	0.7480	0.5349	0.0725	0.6691	0.7754	0.8442	0.1711	0.1630
Decision Tree Classifier	0.6032	0.7110	0.7000	0.7044	0.5529	0.0985	0.6054	0.7130	0.7246	0.0823	0.0822
Gaussian Naive Bayes	0.3587	0.7353	0.0745	0.1341	0.5690	0.0414	0.3505	0.1254	0.0688	0.0155	0.0055

As can be seen from Table 6.1, the highest cross-validation and held-out test F1-scores (CV F1 = 0.8068, Test F1 = 0.8096, Test Accuracy = 0.6887) were attained by Support Vector Machines, whereas the second and third positions went to AdaBoost (CV F1 = 0.8028, Test F1

= 0.8070) and Gradient Boosting (CV F1 = 0.7972, Test F1 = 0.8061), respectively. Nevertheless, closer investigation showed that the good F1-scores of linear and boosting baseline classifiers were largely a result of exploiting the majority class: for instance, classifiers like AdaBoost and SVM had close-to-perfect recall rates (0.9783–1.0000) while having terrible specificity on the minority Class 0 and, thus, obtaining reduced values of Cohen's Kappa (0.0000–0.0888) and rather low MCC scores (0.1115–0.1442). At the same time, non-parametric distance and tree classifiers like KNN and Random Forest exhibited better-balanced classification performance and scored the best values of correlation metrics on the test set (KNN MCC = 0.1711, Cohen's κ = 0.1630; Random Forest MCC = 0.1328, Cohen's κ = 0.1166). Gaussian Naive Bayes failed almost entirely (Test F1 = 0.1254, Test Recall = 0.0688), indicating a serious violation of the conditional independence assumption among the correlated composite features.

Table 6.2: Institutional Cohort Sample Distribution and Binary Target Stratification

Cohort Designation	Constituent Universities	Sample (N)	Class 0 (No/Low)	Class 0 (%)	Class 1 (Med/High)	Class 1 (%)
Overall Cohort	All five participating universities combined	2,036	660	32.42%	1,376	67.58%
Public Universities	DU, JU, CUET	868	284	32.72%	584	67.28%
Private Universities	Daffodil International University, AIUB	1,168	376	32.19%	792	67.81%
Daffodil Cohort	Daffodil International University (Host)	695	211	30.36%	484	69.64%

As can be seen in Table 6.2, the results of institutional comparison show remarkable uniformity of the prevalence rates of anxiety in each type of university. Public Universities had 67.28%, Private Universities had 67.81%, and Daffodil International University sub-sample had 69.64% of anxiety rates. These results provide empirical evidence that AI-induced occupational anxiety is not limited to some particular academic funding models or institutional type but rather it is a prevalent issue within the participating university cohorts represented in the dataset.

6.2 Limitation

An analysis of the computational pipeline, sampling method, and modeling assumptions revealed some limitations of this work:

- **Sampling Concentration at Regional and Institutional Levels:** Despite the fact that the corpus of surveys ($N = 3,156$) is much bigger than typical psychometric anxiety studies [6], [10], [17] in the field of AI adoption, the surveying was done only in the leading universities in urban areas of Bangladesh. In particular, the share of private universities sector, such as Daffodil International University (33.71%) and AIUB (21.96%) was over 55% of total population weight. Therefore, rural colleges, medical institutes [14], [27], polytechnic academies, and non-urban public universities are underrepresented in this sample.
- **Cross-Sectional Observational Design:** Dataset represents one static snapshot in the period of the adoption wave of the ChatGPT generation AI tools (2023–2025) [3], [22]. AI tools capabilities, paradigm of workplace integration, and job market landscape changes over monthly periods. As longitudinal panel data tracking is not used in this work, the research is unable to measure any temporal decay of anxiety or hysteresis effects of adapting and gaining experience in using AI tools [18], [20].
- **Imbalance and Decision Boundary Bias:** The binary target variable has shown a marked imbalance with 67.58% to 32.42% ratio of the positive class (1,376 vs. 660). Standard estimators for unweighted model converged to the heuristic solution that prioritizes sensitivity ($\text{Recall} > 0.95$), thus having high F1-scores. This is confirmed quantitatively by the low Kappa metrics in the case of top-performing classifiers (AdaBoost $\kappa = 0.0000$, SVM $\kappa = 0.0797$), implying that accuracy and F1 metrics mask decision boundary bias.
- **Feature Approximations and Missing Data Imputation:** Some psychological latent constructs were measured through the pragmatic proxy variables instead of the psychometric multi-item scales [4], [27]. Specifically, the construct of Threat_Perception was measured as a binary response variable, obtained through a free-form text completion of the ai_tool_perception item instead of the psychometric measurement. Moreover, ai_knowledge has 1.47% missing values ($n = 30$), which is relatively low but still required median imputation in the training set, thus potentially reducing the variance of the derived Risk_Knowledge_Gap dimension.
- **Lack of Deep Natural Language Processing:** Current design used the regular expressions pattern matching with respect to the list of predefined keywords for detecting tool usage (ChatGPT, Copilot, Midjourney, etc.). While the approach worked well for structured data classification, it ignores the sentiments expressed, explanations about fear of workplace and domain-specific tool combinations found in the unstructured survey narratives [8], [26].
- **Baseline Hyperparameters Limitation:** The current baseline is a preliminary unoptimized model. Classifiers were trained with the default settings in scikit-learn

framework before Bayesian hyperparameters tuning, cost-sensitive decision threshold adjustment, and resampling algorithms (such as SMOTE, Tomek links).

6.3 Future Work

The following specific research efforts are proposed to rectify the above-identified limitations and push the computational frontiers of educational data mining and occupational AI forecasting:

- **Class Balance Fix via Synthetic Oversampling & Cost-Sensitive Loss Functions:** The subsequent implementations of the pipeline will utilize synthetic oversampling techniques such as SMOTE, Borderline-SMOTE, and ADASYN, along with hybrid noise-filtering mechanisms such as SMOTE-Tomek and SMOTE-ENN inside the cross-validation loop in order to synthesize minority-class samples without data leakage. Moreover, cost-sensitive learning techniques, including class weighting and Focal Loss, will be used to provide equal penalties for both types of classification errors.
- **Bayesian Hyperparameters Optimization and Meta-Ensemble Stacking:** Hyperparameter tuning will be carried out with the help of Tree-structured Parzen Estimator (TPE) algorithms using Optuna or scikit-optimize on state-of-the-art gradient boosting machines (LightGBM, XGBoost, CatBoost). In addition, the meta-ensemble of heterogeneous Stacking Classifier will be designed by combining diverse top performers (Random Forest, SVM, LightGBM, Extra Trees) in conjunction with regularized Logistic Regression.
- **Explainable Artificial Intelligence (XAI) and Latent Attribution Modeling:** In order to translate machine learning predictions into actionable institutional intelligence, the interpretability of models will be assessed comprehensively through SHapley Additive exPlanations (TreeSHAP, KernelSHAP) and Local Interpretable Model-agnostic Explanations (LIME). This will enable explicit quantification of the marginal effect of the Risk-Knowledge Gap, Perceived Urgency, and depth of tool usage on the tipping point of crossing the elevated career-anxiety classification threshold for the individual student.
- **Embeddings of Narrative Language Data through Transformer Models:** In order to capture the subtleties of language, the fine-tuned transformer model (BERT, RoBERTa, DeBERTa) will be applied to qualitative narrative features (ai_tool_perception and career_path). The contextualized text embeddings (sentence vectors of size 768) will be added to the current demographic features in the multi-modal tabular-textual setup (TabNet, MLP with cross-attention).
- **Multi-Wave Longitudinal Analysis and Cross-Institutional Expansion:** The survey framework will be extended into a longitudinal panel design to track the same student cohort across subsequent academic semesters and into the first post-graduation year of corporate work experience. The sampling scope will be broadened to include polytechnics, rural degree colleges, and medical academies [14], [27], providing a basis for the comparative analysis of career adaptability and self-efficacy trajectories through Structural Equation Modeling (SEM) [1], [6], [15].
- **Institutional Interventions and Curriculum Design:** Based on the empirical evidence showing the maximum peak of anxiety when the threat literacy gap is the greatest, the administrators and curricula designers of Daffodil International University and other institutions will be provided with the action-oriented policy guidelines. These include

hands-on certification of prompt engineering, ethical AI literacy courses incorporated in the degree curriculum for non-computing programs, and individualized career resilience training aimed at transforming automation dread to adaptability [11], [15], [16].

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